An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

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ABSTRACT

Do nearby stars show artificial narrowband carriers? We searched 405 fine-channel ALMA archival windows toward 61 systems within 40 pc, covering 47.7 GHz in 26 separate intervals between 114 and 873 GHz; 1258 coarser windows constrain only unresolved emission. Of 51 threshold crossings, 23 coincide with catalogued lines, 16 of them with further evidence of line emission. Given the adopted mask and background, the other 28 are consistent with chance (24.0 expected from control positions, $P = 0.24$). No carrier is detected or confirmed: for the crossings, a non-detection, not an exclusion. Coverage, more than depth, limits the search: 20 systems were never observed in two separated sessions and one more only at different frequencies. For a carrier narrower than one channel, ninety per cent single-epoch trigger completeness, not the chance of confirming a signal, needs a median equivalent isotropic radiated power of 1.2×10^{15} W, about 60 times the Arecibo planetary radar's. Continuous carriers above each system's limit, outside the mask and within the searched drift, are excluded at the single-epoch trigger level toward these systems and frequencies. Intermittent or broader signals, faster drifts, beams aimed elsewhere, other frequencies and unobserved stars remain open.

Subject headings: Technosignatures, Search for extraterrestrial intelligence, Submillimeter astronomy, Radio spectroscopy, Debris disks

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Many of those fields contain a nearby star. This paper searches them for a narrowband transmitter at the position of such a star. No such signal was found.

Searches for *technosignatures* – detectable imprints of technology, intended or not (Sheikh 2020; Vidal et al. 2026) – have examined many thousands of stars since the first proposals (Cocconi & Morrison 1959; Drake 1961), from Project Phoenix (Tarter 2001; Backus & Project Phoenix Team 2002) to the targeted, commensal and machine-learning-assisted surveys of today (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024; Morrison et al. 2023; Tremblay & Tingay 2024; Manunza et al. 2025; Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). Even so, the last published accounting of that effort covers only 6×10^{-18} of the space of transmitter properties (Wright et al. 2018), and essentially all of it lies below ∼30 GHz. Above that frequency the record is nearly empty. Steffes & DeBoer (1994) and Mauersberger et al. (1996) observed 40 and 17 nearby targets at 203 GHz, each in no more than 35 MHz of a single line. The union of frequencies searched here is 1364 times that bandwidth, and the coverage of the median system alone, 1.84 GHz, is still 53 times larger. Neither search allowed for drift or tested whether a signal persisted. Of ALMA's receiver bands, only Band 3 has been searched, and that search looked at distant stars in calibrator fields (Mason et al. 2025).

There are good reasons to look at these frequencies. Interstellar dispersion and scintillation are negligible, and an aperture delivers far more gain per watt. The costs are real as well, however. System temperatures are higher, the atmosphere is opaque in places at both ends of the link, and the narrow beam (25 arcsec for the 12 m benchmark of §5.3) would have to be aimed at the Earth deliberately. Since nothing is known about such a transmitter, limits are stated here as an equivalent isotropic radiated power (EIRP), which needs no model of it. The targets were also chosen by other programmes for their own purposes, so the sample cannot bound the fraction of stars that host a transmitter (Wright et al. 2018; Margot et al. 2023; Sheikh et al. 2025), as §2.1 shows.

The signal sought is an *unresolved spectral carrier*: emission confined to a few channels at the stellar position, present at more than one epoch, and allowed to drift in frequency as it must for a transmitter on a rotating, orbiting body (§3.1). Broadband leakage on a planetary scale (Garrett 2026) lies outside the search by construction. An event is worth pursuing only if it is found again (Sheikh et al. 2021), and the survey is built around that test. Recurrence is not enough on its own, however, since a molecular line or a source of interference can also return. Mason et al. (2025) identified the two obstacles particular to this band: drift rates scale with frequency, and the spectrum is crowded with molecular lines. The method therefore searches over drift, compares every event with a line catalogue, and looks for the event again in a separated epoch. As ALMA is an interferometer, the star can also be compared with control positions inside its own primary beam, which a single dish cannot resolve, and that comparison takes the place of the usual interference rejection (§3).

Experiments with different channel widths are compared here on the minimum detectable received flux, $F_{\rm min} = S_{\rm min}\,\delta\nu$, each at its own channel width. The comparison has to be made in this way because $F_{\rm min}$ rises as $\delta\nu^{1/2}$, and a limit holds only for a signal narrower than

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the channel that set it. The channels of the searches compared span 7 orders of magnitude, from 2.79 Hz to 31.25 MHz, while this survey's own run from 15.3 kHz to 31.25 MHz.

Figure 1 draws each published threshold beside its value rescaled to a common 5σ. The rescaling is only an arithmetic convention, since it does not match the false-alarm probabilities of the different searches, and of the limits plotted only this survey's EIRP_{90} is calibrated with injected carriers. The plotted values are in `figures/figure_data.csv` of the data archive (White & Dey 2026).

The closest comparison is with the Band 3 search of Mason et al. (2025), and channel for channel this survey is the shallower of the two except in its deepest windows. At a bare 5σ threshold that search reaches a median $4.9 \times 10^{-23}\,\mathrm{W\,m^{-2}}$ in a 30.52 kHz channel. The deepest window here outside the line mask reaches $1.3 \times 10^{-23}\,\mathrm{W\,m^{-2}}$ in 122.1 kHz, a factor 4 deeper, but the median window reaches only $3.1 \times 10^{-22}\,\mathrm{W\,m^{-2}}$ in 488 kHz, a factor 6 shallower. What distinguishes this survey is instead its frequency range, and its 61 systems within 40 pc, where the Band 3 search observed distant stars in 4 calibrator fields over 117 MHz.

2. THE SAMPLE

This search covers 1663 spectral windows toward 90 stars in 83 systems within 40 pc, where a *system* is a bound group, counted once however many of its stars were observed. The windows come from 407 public ALMA execution blocks, all put through the same pipeline: 4.1 TB of raw data, 302 h of telescope time and 1178 h summed over windows. We call this set the *primary census*. Every expectation in the paper, and every limit of §4.1 and Table F1, is computed over it and nothing else. It holds 51 threshold crossings (Table A1b, row 1.1), all in quality-passing data, one of them toward HD 139084 B, which lies in a census window but has no catalogue row. That crossing is attributed (CO(3–2) at $+24\,\mathrm{km\,s^{-1}}$ in the star's frame). Because its own position has no catalogue row, it stands on neither side of the chance comparison, and its window enters the expectation once, through the control positions around HD 139084 (Appendix D).

Some data were searched after the census was closed, and these are kept apart. A further 2 crossings lie in an *archival tail* of 8 blocks toward 5 census stars, and they enter no expectation. Nor do an archival extension, a reserved *hold-out* (Appendix B.1) and a control beyond 40 pc (§4.4). Table A1 accounts for every block in every set.

The census divides into two experiments by native channel width, since both the search statistic and the completeness calibration depend on it. *Class A*, the primary experiment, is the 405 fine-channel windows toward 66 stars in 61 systems. Of these, 396 are channelised finely enough for the searched drift to cross a channel (§3.1), and every sensitivity and limit quoted is Class A's unless stated otherwise. *Class B*, the other 1258 windows, supports a secondary search for an unresolved narrowband spectral excess. The channelisation of these windows is bimodal, so the division is unambiguous: the widest fine channel is 1.95 MHz and the narrowest coarse one 7.81 MHz, with nothing in between.

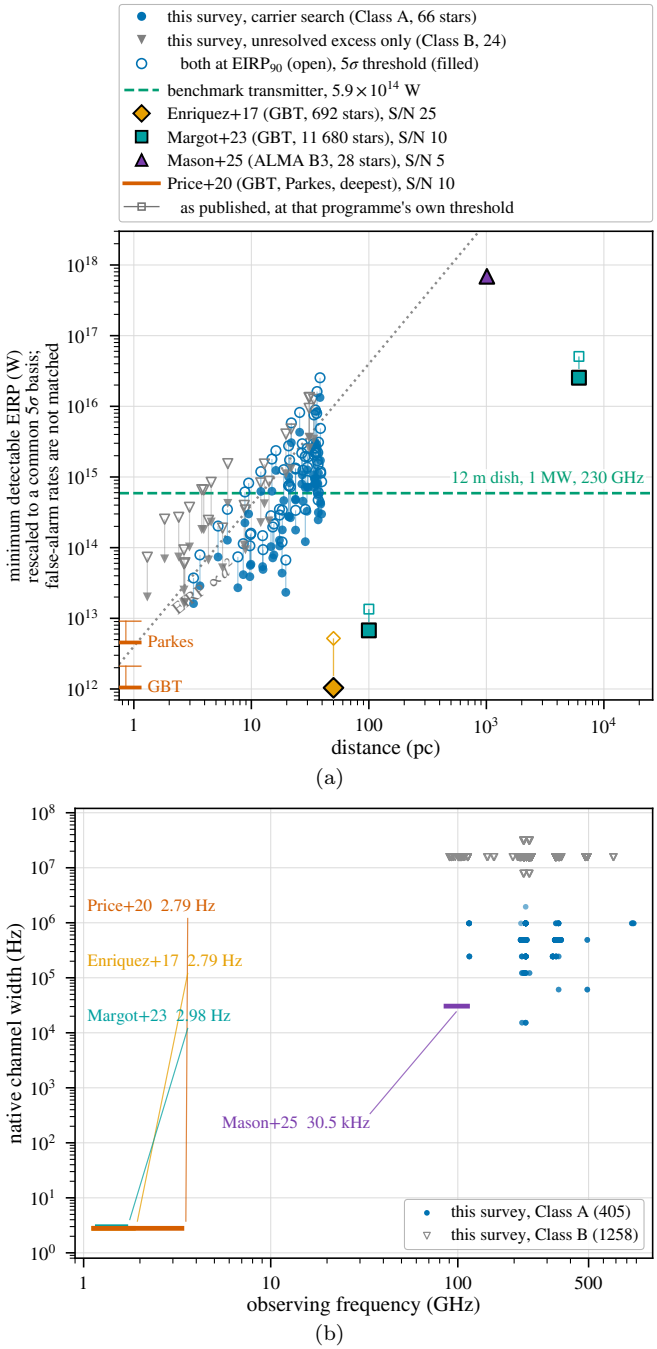


FIG. 1.— Where this survey sits among published radio technosignature searches (§1). **(a)** Minimum detectable EIRP against distance; per star, this survey's deepest carrier-search window (blue) or, with coarse channels only, its deepest coarse one (grey); filled, the bare threshold; open, EIRP_{90}. **(b)** Native channel width against sky frequency (filled, Class A; open, Class B).

A star enters the sample if its Gaia DR3 parallax is at least 25 mas and measured to better than 20 per cent. In practice the second condition never binds, since the least precise parallax among the stars searched has an uncertainty of 2.5 per cent. Its proper-motion-corrected position must also fall inside the quoted field of view of a public execution block, though not necessarily at the pointing centre, and its distance is taken as $1/\varpi$. A window is kept only where the primary-beam response at the star is at least 0.22. At that level the adopted Gaussian beam and a blocked-aperture response part by 10 per cent, and no window falls below it (Appendix A.2). One serendipitous field star is named throughout after the ALMA source at the centre of its field, ALMA J153702653−33192492. It is 2MASS J15370316−3319273 (Gaia DR3 6015967739554000768), an M5 dwarf at 35.0 pc with no catalogued radial velocity.

The Gaia requirement has a cost at the bright end, since a star with no Gaia DR3 parallax cannot enter the sample. It removes all 22 stars in this volume brighter than second magnitude, and 257 of its 4227 Hipparcos stars. Of these, seven (Sirius A, α Centauri A, Vega, Fomalhaut, α Centauri B, β Leonis and MCC 124) lie in public fields, with 38 fine-channel and 374 coarse-channel windows that a search admitting Hipparcos astrometry could add.

2.1. *The archival selection*

This sample was not designed. It is whatever ALMA programmes with other aims happened to observe among the stars Gaia DR3 catalogues, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 1 measures the resulting bias against the volume-limited Gaia DR3 census of which the searched stars are a subset ($\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$, $1/\varpi \leq 40$ pc; 17 566 objects). Both columns are classified by BP−RP colour, with absolute G magnitude separating the degenerates.

Distance is the dominant selection effect. Stars within 5 pc are over-represented by a factor 53, and the excess declines monotonically to a factor of two *under*-representation beyond 30 pc. Colour comes second. A stars are over-represented by a factor of about 17 and F and G stars by factors of three, because those are the stars with bright debris discs. For the A stars that factor is a floor, since the Gaia requirement itself removes stars as blue as these at 3.7 times its rate among the others. M dwarfs, on the other hand, are depleted. They make up 41 of the 80 non-degenerate stars among the 90 searched, a factor 0.7 of their share of the census, and only 28 of them carry fine-channel coverage. Coverage is also concentrated, with 285 of the 1663 windows coming from 3 systems.

The selection acts hardest on the targets a designed search would pick. Of the 16 sample systems within 15 pc with a confirmed planet, 6 have no Class A coverage and so no drift-resolving limit. No statement in this paper is therefore an occurrence rate for the solar neighbourhood.

2.2. *Frequency coverage and archival scope*

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, in 26 disjoint islands between 114.0 and 873.1 GHz rather than a continuous band. With the coarse-channel windows the coverage is 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §3.2. Fig. 2 shows how that coverage is distributed. A transmitter between the islands was never observable here, whatever its power. Nor is the archive exhausted: 177 further blocks inside the same query were unprocessed when the census was frozen, 18 of them because the archive holds their raw visibilities but never pipeline-calibrated them (Table A1).

TABLE 1
Selection function of the 90 searched stars against the volume-limited Gaia DR3 census of §2.1. "Ratio": the searched over the census column fraction.

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	13 (14.4%)	52.9
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	27 (30.0%)	0.5
Colour class[a]			
A	61 (0.4%)	5 (6.2%)	17.2
F	473 (2.8%)	8 (10.0%)	3.6
G	978 (5.8%)	15 (18.8%)	3.2
K	2221 (13.2%)	11 (13.8%)	1.0
M	13 082 (77.8%)	41 (51.2%)	0.7
Other axes			
Confirmed planet hosts[b]	26 (15.5%)	24 (26.7%)	1.7
White dwarfs[a]	463 (2.7%)	8 (9.1%)	3.4
Searched stars / systems	–	90 / 83	–

[a] Colour bins, not MK types: BP−RP < 0.36 for A, 0.36–0.72 for F, 0.72–0.95 for G, 0.95–1.77 for K and > 1.77 for M, where the census's own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K. Degenerate: more than 4 mag below the census main sequence. The five classes are normalised over the 80 non-degenerate searched stars and the 16 815 non-degenerate census objects, the degenerate row over the 88 classified searched stars and the 17 278 classified census objects.

[b] Both columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4. The 24 searched hosts carry 51 confirmed planets, 22 of the 24 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

3. DATA AND METHODOLOGY

A carrier from an accelerating transmitter drifts in sky frequency, and over a long track its power spreads across many channels. The analysis therefore *de-drifts* the data, shifting each integration's spectrum to follow a trial drift rate before adding the integrations together. This section describes that search, how its output is turned into a power, and the three steps through which every event then passes.

3.1. *The de-drifted search statistic*

The search statistic is the largest stacked signal-to-noise ratio anywhere on the channel×drift grid of a window's dynamic spectrum,

$$T(\mathbf{x}) = \max_{\dot\nu,\,\nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot\nu t)\,/\sigma^2(t, \nu + \dot\nu t)}{\left[\sum_t 1/\sigma^2(t, \nu + \dot\nu t)\right]^{1/2}}, \qquad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν, $\dot\nu$ is the trial drift rate, and $\sigma(t, \nu)$ is the noise scale, both evaluated at the channel

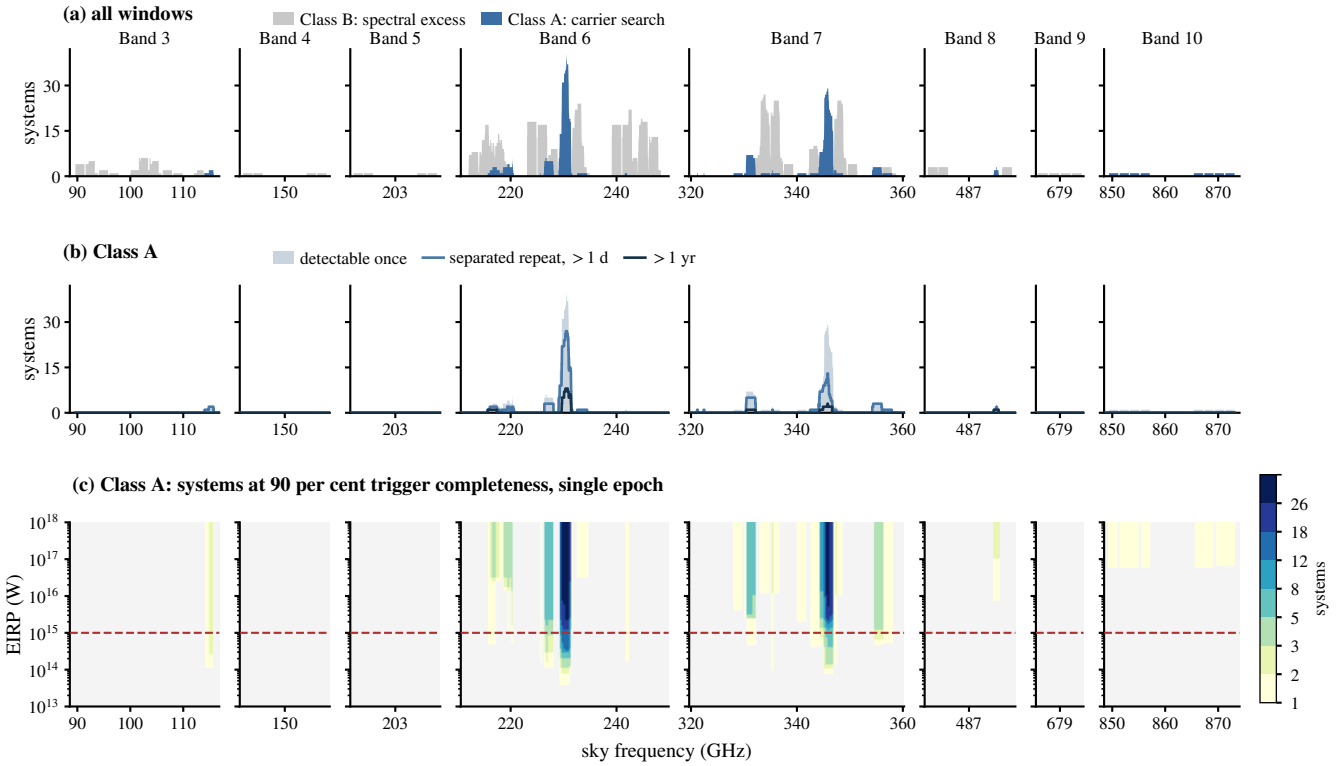


FIG. 2.— **Where this survey searched, and where a repeat could test it**: independent systems covering each sky frequency, one panel per populated ALMA band, its width proportional to the band's frequency share; curves step at the searched window edges. *(a)* Every window: blue, the fine-channel carrier search (Class A); grey, the coarse-channel spectral-excess search (Class B); the summed window bandwidth is 24.1× the union. *(b)* Class A: detectable once (grey); with a *separated* block (§3.2) covering the frequency (blue); beyond a year (dark). An accelerating carrier further needs a predicted frequency within the repeat coverage, with adequate sensitivity. *(c)* Class A systems in which a carrier of that EIRP reaches the threshold nine times in ten in one epoch; dashed, 1×10^{15} W.

the track occupies in that integration. $T_\star$ denotes the value at the star. The time t runs from the first integration, so the frequency that labels a track is the one a carrier would occupy at the start of the block, and a repeat epoch is read at the cell the same convention fixes (§4.4).

Equation 1 is the matched filter, and T an absolute signal-to-noise ratio, only if the noise is independent between integrations and channels. The analysis does not assume that. The trigger is instead set against the distribution T actually takes in the data, measured in every window at 512 control positions on the same visibilities and through the same stack. A value of T is then converted into a power with injected carriers recovered through the unchanged pipeline (§4.1). The controls lie, from a fixed seed, in an annulus set by each block's synthesised and primary beams. The noise scale σ is their median absolute deviation at each integration and channel, scaled to a Gaussian-equivalent standard deviation, and the statistic is evaluated in the same way at the star and at each control. The field read over is the whole primary beam, that is every `FIELD` entry in which the star's primary-beam response reaches 0.05.

Spectra are extracted at the star's apparent position, never from images. For each integration and native channel the pipeline forms the weighted vector average of the phase-shifted visibilities in total intensity, $I = (XX + YY)/2$. That is the dirty-map value at the star, an amplitude and not a deconvolved flux density, so it is not a point-source measurement where the field holds extended emission. The correction for the star's offset from the phase centre is measured with injected carriers (Appendix A.2). The deposited extractor propagates each position to the observation epoch for proper motion and annual parallax. Some products were made by the first forms of the extractor, which propagated proper motion only, and for those windows the parallax loss is corrected window by window in §4.1. Orbital motion is bounded for every component of a multiple in Appendix A.2. The data are the archive's deliveries, most of them recalibrated. When the recalibration was planned, 178 blocks set a quoted limit or held a crossing, and 163 of them were recalibrated with ALMA's final flags. The other 15 kept the delivered calibration, 8 of them then setting a limit and 7 holding a crossing. Some blocks have changed role since, and Table A2 sorts every block by what it holds now. On that count 3 blocks holding a crossing and 8 setting a system's best limit keep the delivered calibration. The table also sets out the 52 blocks of the Data Availability statement, which were chosen later and from every set (`catalogues/recalibration_reconciled.csv`). The change that the recalibration makes at the control positions bounds the chance expectation of this mixed calibration at 23.6–24.2.

Every delivered channel is searched, without regridding, deconvolution or edge trimming. A per-integration median along frequency removes the continuum, and none is taken along time, so no feature narrow against the band can be removed. The delivered channels are smoothed, so a correction computed from the correlator's own channel shape turns flux density times channel width into a radiated power (Appendix A).

The search itself runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, except where a table gives it in the star's frame. Two later steps need the star's own frame: comparing a crossing with the line catalogue, and reading a repeat epoch at the *predicted cell*, which is the single channel and drift rate of the discovery held fixed in the star's frame. An emitter at rest in that frame, as a line is, would be found again there, while a drifting carrier would by then have moved elsewhere. Both steps correct each block's frequency axis for the Earth's motion and then for the star's systemic velocity. That is the SIMBAD radial velocity (Wenger et al. 2000), queried on 2026 October 4, or zero where none exists, as it is, among the stars with a crossing, for ALMA J153702653−33192492 alone. Toward β Pic the value of the ALMA CO analysis of its disc (Matrà et al. 2017) is adopted.

Two power quantities follow from the statistic, and both are defined in Table 2. The trigger level $P_{\rm trig}$ is an internal pipeline level. The sensitivity EIRP_{90} is measured by injecting synthetic carriers of known amplitude into the calibrated visibilities of 49 windows and recovering them through the unmodified pipeline. Every carrier limit quoted is the EIRP_{90} of its own window, given beside its native channel width, except the labelled bare-threshold comparisons of §1, and its error budget is Table 3. The amplitude scale is checked twice. An independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014, which tests the extraction and not ALMA's flux transfer. The quiescent flux density of ϵ Eridani (Burton et al. 2022) is recovered to 1.10 ± 0.31, which measures the scale to 28 per cent. That is consistent with ALMA's own accuracy specification, ±11 per cent for the sample, which we adopt for the absolute scale (Table 3).

The drift range is bounded by an acceleration and not by a frequency, since drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot{\nu}}{\nu} = \frac{a_{\rm los}}{c} \qquad \text{and so} \qquad a_{\rm los} = \frac{c\,\dot{\nu}}{\nu}, \tag{2}$$

and a ceiling on $\dot{\nu}/\nu$ is then the same physical limit in every band. The ceiling is $12.0\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, or $3.60\,\mathrm{m\,s^{-2}}$ by Equation 2, in 401 of the 405 windows, and $13.3\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ ($4.00\,\mathrm{m\,s^{-2}}$) on the 4 TRAPPIST–1 windows, whose innermost planet demands more. The default is roughly the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, and 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend for a general search. The grid is uniform in $\dot{\nu}$ and runs from the negative ceiling to the first trial at or beyond the positive one (Appendix D.5), in steps of half a native channel over the span of the integrations searched. A carrier midway between two trials therefore loses 0.25 of a channel over that span, and at most 0.29 over the longest on-source time.

The survey does not constrain a transmitter whose apparent acceleration exceeds that range, whether it is chirped faster (a chirp is not distinguished from acceleration) or its orbital and rotational accelerations together exceed it. Transiting TRAPPIST-1 b is the most demanding known case. At the default ceiling 29 per cent of its orbital phase would lie out of reach, and the widened grid used on its star matches its maximum acceleration only to within the ±3.1 per cent to which its host's mass and orbit are known.

Searching over drift also only helps if the largest searched drift moves a carrier across about one channel while the star is observed, $\eta_{\rm drift} = |\dot{\nu}_{\rm max}|\tau_{\rm on}/\Delta\nu_{\rm ch} \gtrsim 1$, where $\tau_{\rm on}$ is the time on source. 396 of the 405 Class A windows satisfy it. The other 9 fall below it because their tracks are short, and the per-window catalogue carries $\eta_{\rm drift}$ for every window.

3.2. *From a crossing to a recurrent feature*

Every window passes through three steps in the same order: the trigger, the line attribution and the recurrence tests. They yield in turn a *crossing*, an *unattributed crossing* and, had one been found again, a *recurrent feature*, each defined once in Table 2. A recurrent feature would not yet be an artificial signal, since a line or interference can return as well. Every count below is read from the ledger of every crossing the search produced, and is of primary crossings, one to a window.

The trigger. The level set on $T_\star$ carries no tail probability. T has not been shown to follow any standard distribution under the null hypothesis, and a window holds between 112 and 6.87×10^6 channel×drift cells, so one level buys very different protection in different windows. Every cell that reaches the trigger is recorded, and not only the largest, because a carrier can be hidden as effectively by a noise excursion as by a line.

The background. With no tail probability attached, a fixed threshold is not a fixed false-alarm rate either. The number of crossings expected by chance is instead computed window by window, from the rate at which that window's own control positions exceed the trigger. This assumes that a star behaves like a control position in its own field, which the reserved hold-out blocks test (§4.2.1). The census's 51 crossings, all of them Class A, are the baseline every expectation is set against. Of these, one lies in a window whose control ensemble fails to behave like noise (Appendix D.3), and it is counted but withheld from the amplitude-dependent tests (Appendix D).

The line attribution. At these frequencies many crossings will be molecular lines, so each one is compared with a line catalogue. The comparison is a single database query, made on 2026 October 6 and repeatable as it stands. A crossing within $\pm 50\,\mathrm{km\,s^{-1}}$ of a catalogued transition in the star's rest frame is *attributed*. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond 10 au, with a margin for the systemic velocity. One fixed list of 12 species is searched, whatever a window was tuned to. It was completed at that query, after the crossings had been examined (Appendix C), and the chance comparison of §4.2.1 is therefore conditional on the mask as adopted. Of the 53 crossings of the census and the archival tail, the query attributes 23 and leaves 30. Halving the half-width to $\pm 25\,\mathrm{km\,s^{-1}}$ gives 21 and 32 respectively. An attribution is a coincidence with a catalogue, not an identification or a disposal, so every crossing goes on to the recurrence tests whichever side of the mask it falls. The price of the mask is that the survey constrains nothing inside it.

The recurrence tests. A repeat counts only if it comes from a *separated* block, which here means a block of another observing session and more than 1 day from the discovery. The session matters as much as the interval, because blocks of one session share calibrators, atmosphere, correlator setup and interference. Two different tests are then made, and they answer different questions. The *line-recurrence check* is a same-frequency test. It reads the predicted cell, ± 0.11 to $\pm 0.63\,\mathrm{km\,s^{-1}}$ wide, in a separated block, and so it can find again only an emitter that holds its frequency in the star's frame. The *trajectory comparison* asks instead whether one bounded trajectory inside the drift grid joins two crossings. It is the only test here that a drifting carrier could meet, and while it can separate two crossings it cannot show that they are one carrier. Table 2 states what each test can establish, and Fig. 2(b) shows the frequencies at which a separated block exists at all.

Whether the chain finds what it is meant to find is tested with the injected carriers of §4.1, and with β Pictoris CO, a real feature at a known stellar position, which is recovered and localised in two transitions and 10 execution blocks (§4.3).

4. RESULTS

The primary experiment is the Class A search for a drifting, unresolved carrier: 405 fine-channel windows toward 61 systems, 396 of them channelised finely enough to tell a drifting carrier from a stationary one (§3.1). The secondary Class B search looks for an unresolved spectral excess, with no discrimination by drift, in 1258 coarse windows. Together they make up the 1663 windows of the primary census.

4.1. *Sensitivity*

Every Class A limit in this paper has the same meaning. It is the equivalent isotropic radiated power at which a carrier at the star makes a threshold crossing in nine trials in ten (EIRP_{90}, Table 2), before the spatial rank against the control positions is applied. The Class B limits are built in the same way, but they are limits on an unresolved spectral excess, with no test of drift and with an injection factor of their own (below).

For the median system, on its best window, such a crossing would have come from a carrier of 1.2×10^{15} W. Of the 61 systems, HIP 10679 has no limit at all, because every one of its windows lies inside the line mask, and three more have limits that hold only if their adopted positions are right. The median, and the counts that follow, are therefore taken over the remaining 57, and the three are described at the end of this section. Over these systems the limits run from 3.7×10^{13} to 2.5×10^{16} W (Fig. 3), and 25 of them reach 10^{15} W. A single Class A window is, of course, less deep than the best window of a system, and the median window reaches 2.5×10^{15} W.

Each limit is taken over the channels that the line mask leaves free in the star's own rest frame (§3.2), and this matters most for the finest windows. The observers tuned these to CO, and a 15.26 kHz window spans only 53.9 MHz, which is less than the 76.9 MHz the mask removes about the line. Hence 15 of the Class A windows lie wholly inside the mask and constrain nothing, which is why HIP 10679 has no limit. Using only the 396 windows fine enough to resolve a drift leaves a system's limit unchanged in the median.

The limits come from injection. Carriers were injected at a random sub-channel phase, with a drift drawn from each window's own grid, and recovered through the unchanged pipeline as far as the localisation at the star. In a Class A window without online spectral averaging, nine in ten come back at $2.74\,P_{\mathrm{trig}}$, where P_{trig} is the bare 5σ of the window itself. With two-channel averaging the figure is $1.90\,P_{\mathrm{trig}}$. That power is EIRP_{90}.

The class-average factor of $\times 2.99$ in Table 3 is related to these two, but it is never used on its own. It was measured with the channel profile the injections deposited, and each window's own response, measured on identical carriers, changes it by $\times 0.91$ without averaging and by $\times 0.64$ with two-channel averaging. The limits use the window's own factor, so $\times 2.99$ is never applied to a window as it stands. The factor shows no trend with the brightness of the control ring, so one value can serve the whole class, while each window's own channel response is carried by injections into the extraction that the search used (Appendix A). Class B has its own factor, $3.65\,P_{\mathrm{trig}}$ (1.71 with two-channel averaging), measured with 5 936 carriers injected into 35 coarse windows.

The calibration of the class factor is uncertain by $\times 0.89$–$\times 1.11$ (Table 3). This interval applies to the median and to the counts, which are taken over many windows. It places the median system between 1.1×10^{15} and 1.4×10^{15} W, so the second figure of the median is set by the calibration.

Some windows already hold something above the trigger at the star, and their limits are treated differently. This moves the best-window limit of four systems, AU Mic, GJ 3379, LHS 1140 and CP−72 2713 (marked $^{\parallel}$ in Table F1), and leaves the median system limit and the count of systems at 10^{15} W unchanged. There are 30 such Class A windows, the 28 that hold an unattributed crossing and two more where only a weaker cell, behind an attributed crossing, is unattributed. In them a carrier that only just reached the trigger would not have stood out from what is already present, so we quote instead the power at which nine injected carriers in ten reach the window's own statistic, which lies between $T_\star = 5.02$ and 5.95. None of these windows was itself injected, so the figure is read from the pooled injections at that level, and it raises their limits by at most 16 per cent. Quoted in this way, these limits are upper limits as well.

These limits describe the trigger completeness of the search, that is, the chance that a carrier at the star makes a threshold crossing. The spatial rank against the 512 control positions is not charged against them, because it decides only which crossings are examined first. A crossing that fails the rank stays in the census, counts against the chance expectation and goes on to the recurrence tests like any other (§3.2).

Charging the rank would raise the limits, though not by a single factor. Among the injected windows the cost runs from $\times 1.00$ to $\times 2.40$, with a median of $\times 1.16$, and it grows with the brightness of the control ring. Averaged over the class it is $\times 1.27$, which would move the median system to 1.6×10^{15} W. If each window is instead charged at the cost measured for windows with a similar ring, the median becomes 1.5×10^{15} W and 23 systems still reach 10^{15} W. In 3 windows with a very bright ring, two

TABLE 2
Operational definitions, used in this sense throughout. A crossing is a single-epoch event; a recurrent feature needs a second, separated epoch, which no sensitivity supplies, and is still not by itself an artificial signal.

Term	Definition
Crossing	A channel × drift cell at the stellar position reaching $T_\star \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its *primary* crossing, any other a *secondary* one. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing more than $50\,\mathrm{km\,s^{-1}}$ in the star's frame from every transition of the line catalogue. A target for follow-up; not a detection.
Recurrent feature	The same event found again in a separated block (§3.2), either at the cell the discovery predicts in the star's frame (the *line-recurrence check*) or in an additional separated epoch at the frequency an independently specified trajectory predicts, with adequate sensitivity there. Recurrence is necessary for an artificial origin but not sufficient, since line identity, interference and the behaviour of the frequency in the star's frame must also be checked and set aside. The check recovers only an emitter at rest in the star's frame, as a line is, so a null there never excludes a drifting or re-tuned carrier. A pair of crossings that one bounded trajectory inside the drift grid joins is *open*, not recurrent: the joining test can reject a pair and cannot accept one (Table 4). For 21 Class A systems this archive holds no separated block at the frequency, so the line-recurrence check can confirm nothing there, whatever the sensitivity.
P_{trig}	$4\pi d^2 S\,\Delta\nu$, the power at distance d of a carrier that makes $T_\star$ reach $T_{\mathrm{trig}} = 5$ in one native channel of width $\Delta\nu$: an internal pipeline level, not a sensitivity.
EIRP_{90}	The power at which nine injected carriers in ten produce a crossing at the star (in a window holding an unattributed crossing, nine in ten reach that crossing's own statistic; §4.1), before the spatial rank against the control positions is applied. This is what the paper means by a carrier limit. The Class B limits of §4.5 bound an unresolved excess and carry their own injection factor. EIRP_{90} is a measured single-epoch trigger completeness and should not be read as a detection probability. One class factor, calibrated by injection in 49 Class A windows and multiplied by the measured response ratio of each window's averaging mode, sets every Class A limit. It is interpolated within the injected configurations for 299 windows and extrapolated beyond them for 57 (§4.1); each injected window's own point (×0.81–×1.12) is in the data archive. It excludes atmospheric decorrelation (Table 3).
Class A	The primary experiment, defined by channel width: 405 fine-channel windows, median 488.3 kHz, toward 61 systems over 47.7 GHz of union bandwidth, 396 of which satisfy $\eta_{\mathrm{drift}} \gtrsim 1$ (§3.1).
Class B	The secondary experiment: 1258 coarse-channel windows, mode 15.625 MHz, constraining an unresolved narrowband spectral excess. Never a carrier search: in 1218 of them no searched drift leaves one channel, and in the other 40 the whole searched range moves a carrier 1.02–1.43 channels, which registers one but does not discriminate it.

toward β Pic and one toward HD 48370, the rank could not be met at any injected power. The best windows of HD 139084 and HD 48370 have a ring as bright, so the rank cannot be charged on either. HD 48370's is one of those three windows, while HD 139084's was not injected. Counting each at its next-best window that can be charged, or as having no limit where it has none (HD 139084), leaves the median and the count just given unchanged.

The largest caveat on these limits is that most of them are transferred rather than measured. Carriers were injected directly into 49 of the 405 Class A windows, 12 per cent, and the result was carried across to the other 356. The injected windows were whole execution blocks, drawn at random and balanced by band and array, and they resemble the population well. Their medians match to within 7.1 per cent on five of the 6 properties compared, and differ by ×1.9 on the sixth, the noise level (Appendix F). Only the ninety per cent point, in units of each window's own trigger power, is carried across. Among the injected windows that point ranges from ×0.81 to ×1.12 of the adopted factor, with 40 of the 49 within a tenth of it. This dispersion, and not the calibration interval, measures how uncertain the limit of one uninjected window is.

Most of the transferred windows, 299 of the 356, lie inside the injected range on all five of the properties that set the reading. The other 57 windows lie outside it on at least one, where that range is not an empirical bound. In all, 6 systems take their headline limit from one of them, and none of them is among the 25 reaching 10^{15} W. Only 7 systems had their best window injected directly, and these are marked in Table F1. For the others the headline limit is an estimate carrying that dispersion, and so is the median, except that the dispersion was not measured where the transfer is an extrapolation. Leaving out the 6 systems concerned (marked with a pilcrow in Table F1), the median over the other 51 is 1.1×10^{15} W, and 25 of them reach 10^{15} W.

We have tried to see how much this matters. Drawing each transferred window's factor at random from the measured ones, once per execution block over 16 000 trials, lowers the median by 2 per cent and leaves 25–26 systems at 10^{15} W in nine trials in ten. Even applied at its extreme to every transferred window, the spread could move 3 systems across that level, 1 out and 2 in. Each system's probability of crossing it is given in the data archive.

Two further effects lie outside the injections. The first is atmospheric decorrelation, which a real carrier suffers and an injected one does not, so every limit is nominal in this respect. Estimated from each block's phase-calibrator phase rms (Cortes et al. 2026, §§10.4.11, 11.3.1) with the phase-scatter relation of Richards et al. (2022, Eq. 3), it makes the median system's best window optimistic by 3 per cent. For most systems this is small, but it exceeds a tenth for 11 of them and reaches 200 per cent for HD 172555. Divided out, it would raise the median limit from 1.2×10^{15} to 1.3×10^{15} W, and would leave 25 systems at 10^{15} W. The uncorrected val-

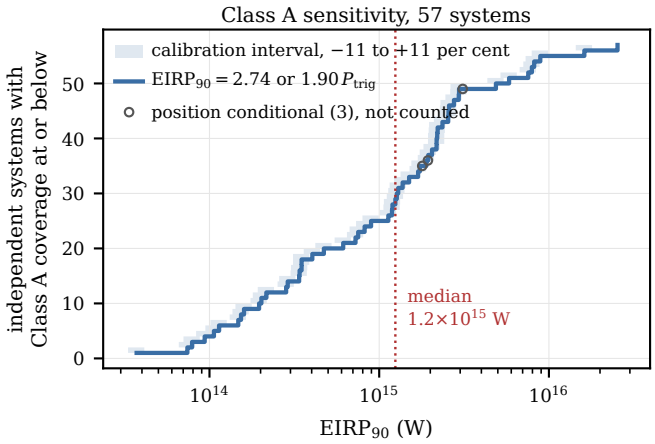


FIG. 3.— **Single-epoch trigger completeness of the carrier search**: the cumulative number of systems whose best window reaches a given EIRP_{90}, decorrelation excluded, over the 57 of the 61 systems with a limit and a bounded position (§4.1), the calibration interval of Table 3 shaded. No power buys a separated repeat at the same frequency, which 21 systems lack (Fig. 2b; §5.3).

ues remain the adopted ones, because only 279 of the 407 blocks record the phase rms, and the others, estimated from blocks matched in baseline, are measurably the harder ones. Table F1 gives the loss system by system, with the corrected limit for those that lose more than a tenth, and the adjusted limits are in the data archive. The channel response, online spectral averaging included, is by contrast measured and applied.

The second effect is the position of the star itself. For Barta 161 12, HD 155555C and LHS 1140 the astrometric solutions are disturbed, and there is neither a companion nor a Hipparcos epoch with which to correct them (Appendix A), so the amplitude lost at the adopted position has no bound. Their limits in Table F1 hold only if that position is right, and these three systems are left out of the median, the count at 10^{15} W and Fig. 3.

A carrier that is not always radiating is penalised in proportion. In 3 888 undrifted trials down to a 10 per cent duty cycle the recovered amplitude followed the duty cycle to within 1 per cent, so a transmitter that is on for a tenth of the time would need 10× the power quoted. The drifting, intermittent case has not been measured.

Two of the nearest stars appear only in the secondary search. Barnard's Star and Wolf 359, the second- and third-nearest stellar systems once brown dwarfs are set aside, were not targets of the 203 GHz searches of Steffes & DeBoer (1994) and Mauersberger et al. (1996), and have otherwise been searched at centimetre wavelengths (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). The archive holds only coarse Band 6 windows toward them, so their values, 2.5×10^{14}–3.4×10^{14} W over 12 windows and 2.7×10^{14}–3.9×10^{14} W over 12, are limits on an unresolved narrowband spectral excess, and to our knowledge the first at millimetre or submillimetre wavelengths for either star.

No Class B window that passes the quality criterion holds a crossing, which is what chance predicts. Computed exactly as for Class A, the 1254 windows expect 0.12 crossings from their own control positions ($P(0) = 0.89$). The expectation is small because a coarse window holds a median of only 256 independent cells, against 193 200 in Class A.

4.2. *Threshold crossings*

Of the 1663 windows of the primary census, 51 hold a cell reaching $T \geq 5$ at the stellar position, all in data that pass the quality criterion of Appendix D.3. The window flag of that appendix reaches 5 windows in 2 blocks, and the one crossing it reaches, toward HD 48370, is withheld from the tests that depend on amplitude. The $\pm 50\,\mathrm{km\,s^{-1}}$ line mask places 23 of the 51 within that velocity of a catalogued transition in the star's own rest frame (Appendix C). Those 23 are attributed, 16 with corroborating evidence of line emission and 7 on the coincidence alone, and 28 are unattributed. None was identified as interference by the available diagnostics (Appendix E). The archival tail adds two more, both unattributed and both toward HD 14055, in windows that no expectation covers (Table A1). They are in the ledger, but in none of the counts above. The correlator's frame suggests that the attributions reflect line emission, and not merely the width of the mask, since in that frame the crossings crowd onto the transitions these setups were tuned to (Appendix E).

Of the 51, 11 outrank all 512 control positions of their own window. Of these, 10 are carbon monoxide toward β Pic and HD 48370, and the other, toward 61 Vir, is the only unattributed crossing to do so. It is not present again at its own frequency in the star's frame (§4.4). Behind the windows' largest cells, twenty-three weaker cells in all also reach the trigger. Of these, six are unattributed, and none reaches the trigger in any covering block the recurrence test reads. None of the unattributed crossings is found again as a recurrent feature (§4.4), and this survey reports no detection; the deposited crossing ledger lists every one.

4.2.1. *How many crossings chance predicts*

We first ask whether the unattributed crossings are more numerous than noise alone would give. The number of searched channel × drift cells in a window spans 4.8 decades, so the expectation is built window by window over the 405 Class A windows of the census, from each field's own control positions. The same condition is applied on both sides: a cell at the position that reaches the trigger and that the line mask does not attribute (Appendix D). The control positions predict 24.0 unattributed Class A crossings, and 28 are found. A count that large or larger arises by chance with probability 0.24.

The expectation uses one share of control exceedances attributed by the mask, pooled over all the windows, and measuring that share instead for the windows above and below the median masked fraction (24.1 expected, $P = 0.25$) or window by window (22.1, $P = 0.14$) leaves the conclusion as it is, so the pooled share is kept (Appendix D).

That probability, and the others built from the control positions here and in §4.4, rest on two conditions. The first is that the control positions are a fair model of the background at the stellar position. The searched data cannot test this. A residual of calibration or imaging that raised the rate of crossings at the star without raising it at the controls would change the probabilities, and a calibration error that varied with frequency is one such residual, since it would raise the star and not its

TABLE 3
What sets $EIRP_{90}$, and in which direction. The calibration interval applies to the class-average factor and so to statistics over many windows, and the spread between windows is the statement about one window. Decorrelation is a one-sided bias. The retentions applied window by window are listed below the table (§4.1).

	value	what it does to the limit
Class-average completeness	×2.99	nine carriers in ten recovered at this multiple of a window's own trigger power, with the deposited channel profile. *Measured* in the 49 Class A windows injected directly and *estimated*, at this factor, in the other 356
Own channel response	×0.91, ×0.64	the ninety per cent point with the window's own response, online spectral averaging included, over that with the deposited profile, without and with two-channel averaging ($N = 1$, $N = 2$; ×0.68 at $N = 4$). *Measured* on identical carriers in 24 Class A windows and *applied*: a window's factor is ×2.74 or ×1.90
Calibration interval	−11/ + 11 %	×0.89–×1.11 on the factor, and so on the median limit and the system counts. Combined in quadrature from the factor's sampling error over the 49 injected windows (±2 per cent), the observatory's band-dependent flux-scale specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ; ±5 in Bands 1–5; ±10 in Bands 6 and 7; ±20 in Bands 8–10, ±10.2 per cent weighted over this sample's bands) and the distance and visibility scale (±4.5 per cent); ±21 per cent for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
Window to window	×0.81–×1.12	an observed dispersion, not an error bar: the 49 injected windows resolve 2.41–3.36 $P_{\rm trig}$ individually, 40 of them within a tenth of the pooled value. This, and not the interval above, is what an estimated limit is uncertain by; it is wider. It applies to the 45 transferred limits inside the injected configurations, not to the 6 outside them
Decorrelation	+0/ + 200 %	stated and not applied: measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does. One-sided, and a floor. A median +3 on the systems' best windows, +8 above 400 GHz, and more than +20 in 5 per cent of windows. Only 279 of the 407 blocks record the phase rms

Applied window by window, and so not part of the budget (Appendix A): the intra-integration smearing, ×1.00–×1.05, from each window's own integration length where it is recorded (1098 windows) and bounded where it is not (565); the primary-beam response at the star's offset, ×1.00–×4.5, with a one-sided model departure of up to 7.1 per cent in the 12 windows furthest off-axis; and the annual parallax, ×1.00–×2.06, only where a product predates the extractor that centres it on the star's apparent position (57 Class A windows; bounded on the 13 with no parallax record).

continuum-free controls. To make a crossing on its own it would need 21.4 per cent of the star's continuum even in the least demanding window, but a smaller residual adding to the noise could still change the rate. As a sensitivity check, the stellar rate can instead be calibrated at the ratio 1.102 that the star shows over its controls just below the trigger, and the expectation is then 26.4, at $P = 0.41$.

The second condition is the mask itself. Its species list was completed after the crossings had been examined, and SO was added after a crossing had been found on one of its lines (Appendix C). The probabilities, and the allowance given below, are therefore conditional on the mask as adopted. Without SO the same comparison gives 31 observed against 24.9 expected, at $P = 0.14$.

The expectation is not carried by a few noisy fields. Without the ten windows whose control positions trigger most often, the remaining 395 predict 24.0 against 28 observed. In 12 of 12 subsets by band, array, channel width and field type, the observed count lies within the central 95 per cent of its own predicted distribution.

Both sides of the comparison are read over the whole primary beam, the field of §3.1. The extraction's original rule was narrower, keeping only the one field entry whose phase centre lies nearest the star, which drops on-source time and real signal. Neither rule has an adjustable radius. Read over that narrower field on both sides, the comparison gives 31 against 23.3, at $P = 0.08$. Neither geometry was nominated as the primary before the counts were seen.

Because the field and the mask were both chosen after the crossings were seen, the comparison was repeated for all eight combinations of field, mask and background rate (Table D1, Appendix D.2). The smallest probability among them is 0.042, for the nearest field without SO, and once allowance is made for its having been picked from the four combinations of field and mask it becomes 0.17, so no combination departs from chance at the 5 per cent level.

The comparison can also be read the other way, as a measure of how much larger the background could be before the observed count became improbable. If 14 is added to the mean number of crossings expected from the control positions, a count as low as 28 still has a probability of 0.06, above the one-sided 5 per cent level, while one more brings it to 0.04, below that level. On the narrower reading the figure is 18. This is a statement about the mean of the background model, and it carries the same two conditions as the probability. It is not a confidence bound on how many of the observed crossings are real, and still less a count of transmitting systems or an occurrence rate (Appendix D).

Stellar flares are excluded by a broadband prediction. A millimetre flare raises every channel of a window alike, so one able to make a single-channel crossing would appear as a 53–307σ continuum event in the same integrations. The observed continuum statistic of the crossings, by contrast, has a median of −0.29, and the AU Mic flare of MacGregor et al. (2020), in a block that also holds a crossing, adds only 0.17σ to the single-channel statistic.

4.3. *β Pictoris: the positive control*

A search that finds nothing must show that it could have found something, and β Pictoris provides that test. Without being told the disc is there, the pipeline recovers its circumstellar carbon monoxide (Dent et al. 2014; Matrà et al. 2017) in two rotational transitions, in 10 execution blocks drawn from 6 observing sessions over 8.4 yr. In the star's rest frame, at each block's own epoch, all 10 crossings lie within 0.5 km s^{-1} of the adopted systemic velocity. That velocity, however, was itself measured from

ALMA observations of the same CO (Matrà et al. 2017), so the agreement checks the registration from one epoch to the next, and it cannot check the absolute frame. The 10 velocities have a standard deviation of $0.3\,\mathrm{km\,s^{-1}}$ and a full spread of $1.0\,\mathrm{km\,s^{-1}}$. The same emission is the positive control of the line-recurrence check of §4.4: every feature recurs in another execution block, and each transition in another session. Emission of this kind is what the line mask removes, including the CO(2→1) toward HD 48370 that Cataldi et al. (2023) report and flag as possible cloud contamination (Appendix C). The mask covers 12 species and costs 6.4 per cent of the Class A union bandwidth, and it is the single most productive test in the chain (§5.3).

4.4. *Candidates and recurrence*

The field leader. One unattributed crossing has a stellar statistic above every one of the 512 control positions in its own window: the one toward 61 Vir at 346.3232 GHz, in Band 7 and in the star's own rest frame (Table 5; Fig. 4). The rank decides which crossings are examined first, and on its own it disposes of nothing (§3.2). If the stellar and control positions were exchangeable, the control ensembles would predict 0.50 unattributed census crossings leading their field by chance, and one or more with probability 0.40. In the reserved blocks 0 of 6 crossings lead their field, and beyond 40 pc 0 of 6. A visibility-domain fit was made for 38 of the 51 census crossings, and for 32 of them it is consistent with a point source at the star. It does not reject this one, and injected point sources in the same bin return $\mathrm{Re}/\sigma = 6.58 \pm 1.97$.

The star has been observed again at the same stellar-frame frequency in a separated session, and the two separated blocks, the nearest 24.12 h from the discovery, hold a line-like emitter returning to the predicted cell below 0.67 of the discovery amplitude. The carrier's own fitted drift would take it out of that half-channel cell ($\pm 0.211\,\mathrm{km\,s^{-1}}$) in 77 s, so the repeat reading is the line-recurrence check of Table 2. At the nominal threshold-equivalent amplitude $5S/T_\star$ the separated blocks would recover such an emitter with calculated probability 0.97, conditional on that amplitude, and injected carriers measure 0.94. In Table 5, T_{rep} is one cell at one trial and may be negative, and the recovery chance in brackets, at the discovery amplitude, is biased high. The readings behind each bound are in `candidates/repeat_epoch_readings.csv` of the data archive (White & Dey 2026).

The predicted cell. The same check, applied to every crossing, reads the one cell each discovery predicts at its nearest separated repeat. It works on lines, since 13 of the 19 attributed crossings tested recur in another session, 39–1984 d away, and the 6 that do not are the faintest of them. On the 27 unattributed crossings whose cell could be read it returns nothing. The largest readings are 3.78 at the carried drift and 4.71 at any drift, against a trigger of 5. Of the 27, 25 hold a line-like emitter returning to its cell below the discovery amplitude, to a median 0.40 of it. At the weaker nominal threshold-equivalent amplitude, 6 of the 27 are too shallow to test. A drifting carrier, on the other hand, would not stay in that cell, so this null speaks only to an emitter that holds a stationary frequency in the star's frame. The bounded crossings accelerate at 0.66 to $3.55\,\mathrm{m\,s^{-2}}$, or 0.63 to $3.52\,\mathrm{m\,s^{-2}}$ in the star's frame. At that rate a carrier leaves its half-channel cell in 60 to 471 s, and each crossing's nearest covering block lies at least 221 times that residence time away. Nor can the check be turned into a test of a drifting carrier by carrying the fitted drift forward, since the drift's own resolution moves the predicted cell by a median 23 channels by the nearest repeat. A drifting carrier is tested instead by the widened window and the pair test below. No bound depends on the cell's registration, the anchor or the threshold (Appendices B.2 and D.4). Three further unattributed crossings are not tested: β Pic at 241.77 GHz has no separated repeat block, and no other block covers HD 61005 at 856.79 and 871.09 GHz. None of the unattributed crossings is therefore a recurrent feature in the sense of Table 2, which is only the first of the requirements that table sets for an artificial origin.

The widened window. A transmitter in orbit changes its line-of-sight velocity between epochs, so the broadest reading asks whether anything reaches the trigger, at any drift, anywhere in a covering window of a separated session. In the median case that window is $\pm 801\,\mathrm{km\,s^{-1}}$ wide. Of the 96 distinct covering windows, 11 hold a statistic above the trigger, against the 8.6 that each window's own control positions predict ($P = 0.24$; Appendix D). Every one of those windows holds a crossing the ledger already lists, so the comparison re-reads the trigger-level count of §4.2.1 rather than adding a recurrence to it. Of the 11, 8 are toward ALMA J153702653-33192492 and HD 14055, which are also the two most heavily observed objects in the survey: they hold 20 per cent of the Class A census and 49 of the 96 covering windows, so 5.7 of the 8.6 expected exceedances are theirs before anything is looked at. Over the remaining 47 windows the reading is 3 against 2.9, at $P = 0.56$.

The pair test. A carrier that moves might also be caught as two crossings toward one star, and Table 4 asks of the 9 pairs inside the widened window, across 4 stars, whether one transmitter could have produced both. Constant line-of-sight acceleration would require equal drift rates, and the fitted rates differ by more than the grid's resolution in 9 of the 9 pairs, so the question becomes a weaker one: whether any trajectory inside the searched grid joins the two. It separates 1 of the 9. Of the 8 it cannot separate, all are *open* and none is *edge-limited*. An open pair is not a confirmation, since the comparison can reject an association but cannot accept one (§3.2). Chance alone, at each census window's own control rate, would put 3.8–3.9 open pairs in separated sessions against the 4 observed, with the pairs that have an archival-tail member set aside. Of random pairs, 91–94 per cent are joinable. As a test of a drifting carrier it is therefore incomplete. Counting archival-tail blocks, one star, HD 14055, has three separated crossings that one trajectory joins pairwise, and in the census alone none does. However, no constant acceleration or circular orbit inside the grid joins all three. Predicted from the first two, the third misses a constant acceleration by at least 1139 resolutions, and the best circular orbit through all three misses by 56. A line that does not drift behaves as it should under the same test: the CO(2–1) line of HD 285968 is joined across 5.0 d to within 0.6 of the resolution. The accounting for each pair is `candidates/`

`drift_pair_accounting.csv` of the data archive.

HD 14055 and ALMA J153702653−33192492. HD 14055 holds 4 of the exceedances against 4.7 expected ($P = 0.71$), so its concentration is its exposure. Its fields are ordinary, with control rings 1.04 times their class median against 1.00 for the survey. Each of its 19 census blocks pairs two 15.6 MHz continuum windows with two 488 kHz windows, and all 6 of its crossings lie in fine windows that hold the CO(3–2) and ^{13}CO(3–2) lines, 123–1069 km s^{-1} from the nearest masked transition in its own frame. Two of its crossings were followed up in depth, by blocks outside the census. For the first, the 9 follow-up blocks in separated sessions read -0.7 ± 0.7 mJy for a line-like emitter at the predicted cell, against a discovery amplitude of 8.0 ± 1.6 mJy, which bounds it at 0.37 of the discovery. For the second, 6 such blocks read 0.6 ± 1.0 mJy against 10.2 ± 1.9 mJy, a bound of 0.57.

ALMA J153702653−33192492 is a serendipitous dwarf 5.75 arcsec from the phase centre of fields pointed at the Lupus clouds, observed in 212 s snapshots. It holds 4 exceedances where 1.0 were expected, with $P = 0.013$ for this object alone. That probability, however, belongs to an object picked out because it is the most discrepant of the 16 objects compared, and in Poisson-binomial counts simulated at each object's own per-window control rates some object is that far out in 7 per cent of realisations. Its four exceedances lie on four crossings between 216 and 220 GHz, in two of this object's seven blocks, and their control rings are ordinary. Their fitted drifts lie 15–35 grid steps from zero, against an uncertainty measured by injection of at most 0.7 step. Because the snapshots are spread over a 3613 s track, $\eta_{\rm drift}$ is 19 over the track and 1.13–1.14 over the time on source. A visibility fit places two of the four at the star. A wider list of 225 species places a transition within 15 km s^{-1} of every crossing toward either object, as its 24 976 transitions guarantee, and so identifies nothing. We have no explanation for these crossings. ALMA J153702653−33192492 is a target for follow-up, not evidence of a non-astrophysical source, and a repeat observation should point at it, with a radial velocity and one deeper epoch at 216–220 GHz.

Blocks outside the census. Three sets of execution blocks enter no census statistic (Table A1), and each was searched with the same pipeline. The reserved holdout, set aside by the identifier rule of Appendix B.1, has 315 windows toward 36 stars and holds 6 crossings, the strongest at $T_\star = 8.64$. Of the 5 read at their predicted cells, only a molecular line is present again ($T_{\rm rep} = +8.47$). Compared with chance in the same way, its 62 fine-channel windows expect 4.3 unattributed crossings and hold 5 ($P = 0.42$). Its crossings were listed in the same session as the species-list query of 2026 October 6, so the record cannot tell whether they were seen before the mask was completed, but no unattributed one lies within 223 km s^{-1} of a listed transition, and the later decision on SO changes none of them (Appendix B). The archival extension's 463 windows in 115 blocks hold 10 crossings, 7 of them unattributed against 4.5 predicted by chance ($P = 0.17$). The control sample beyond 40 pc holds 6 crossings. Read in the sky frame it supports no limit, and shows only that the pipeline behaves in the same way beyond the sample.

TABLE 4
Can two crossings be one carrier? The 9 pairs toward one star with an unattributed member, and each HD 14055 triple (§4.4).

Star	blocks	new sess.	Δt (d)	$\dot\nu_{1\star}$ (Hz s^{-1})	$\dot\nu_{2\star}$	$\Delta\nu$ (MHz)	miss	one? grid
61 Vir	X822/X82f	yes	3.00	+1092	+3221	+755.4	15	open
J1537−3319	X10359/X398	yes	123	−2000	+2390	+223.5	3.6	open
CP−72 2713	X4a6d/X502d	no	0.09	−2044	+995	+495.6	1476	no
HD 14055	X24be0/Xd46	yes	1.92	+1665	−3823	+512.7	100	open
	X2b2ee/X3835	yes	2.09	−2390	+2155	+395.5	55	open
	X3835/Xd526	yes	607	+2155	−2563	−640.4	–	open
	X2b2ee/Xd526	yes	609	−2390	−2563	−244.9	–	open
	X3835/X25c5e	yes	649	+2155	+2611	+338.3	–	open
	X2b2ee/X25c5e	yes	651	−2390	+2611	+733.8	–	open

Triple	constant acceleration	circular orbit
X2b2ee/X3835/Xd526	misses by 1148	misses by 133
X2b2ee/X3835/X25c5e	misses by 1139	misses by 56

New sess.: *yes* where the two blocks belong to different observing sessions. In the 61 Vir pair, block X82f holds the field leader of Table 5. Starred rates are in the star's frame, the observatory's own acceleration removed (observatory frame: Appendix D.5). *Miss*: departure from constant acceleration, in units of the drift grid's resolution over Δt; a dash, a pair that resolution cannot separate. *One?*: whether one carrier could make both crossings. A pair is *yes* when its miss is 3 or less, which is consistent with one carrier but does not confirm one, and *open* when it misses by more, or cannot be separated, but some bounded trajectory inside the drift grid still joins the two. A pair that no such trajectory joins is *no*, an *edge* pair has one rate on the grid's outermost trial, which is then a lower bound, and an *untest.* pair has no block of another session covering the window about it. The last rows give each triple's miss of its third crossing, predicted from the first two, in resolutions.

TABLE 5
The field leader, the crossing whose stellar statistic exceeds all 512 control positions of its own window (§4.4).

	61 Vir
Frequency, stellar frame (GHz)	346.323236
Drift rate, observatory frame (Hz s^{-1})	+3182
in the star's frame	+3221
$T_\star$	5.947
Highest control T	5.58
Fitted amplitude (mJy)	118.6 ± 19.9
Peak-channel EIRP (W)	5.0×10^{14}
Repeat epochs measured	2
Nearest separated epoch, separation (h)	−24.12
T expected if it stayed in the cell, $T_{\rm pers}$	8.14
T measured, matched drift, $T_{\rm rep}$	+0.42
Amplitude still bounded, $(T_{\rm rep}+5)/T_{\rm pers}$	0.67 S
Recovery chance Φ if the amplitude is $5S/T_\star$ (S)	0.97 (1.00)
Nearest masked transition	SO(9_8–8_7)
its offset $\Delta v_\star$ (km s^{-1})	−177.6
Disposition	not confirmed

Notes. S is the amplitude of the discovery crossing. The recovery chance Φ is given for a carrier of amplitude $5\,S/T_\star$, at which the discovery would only just have reached the threshold, and in parentheses for one at the full discovery amplitude S.

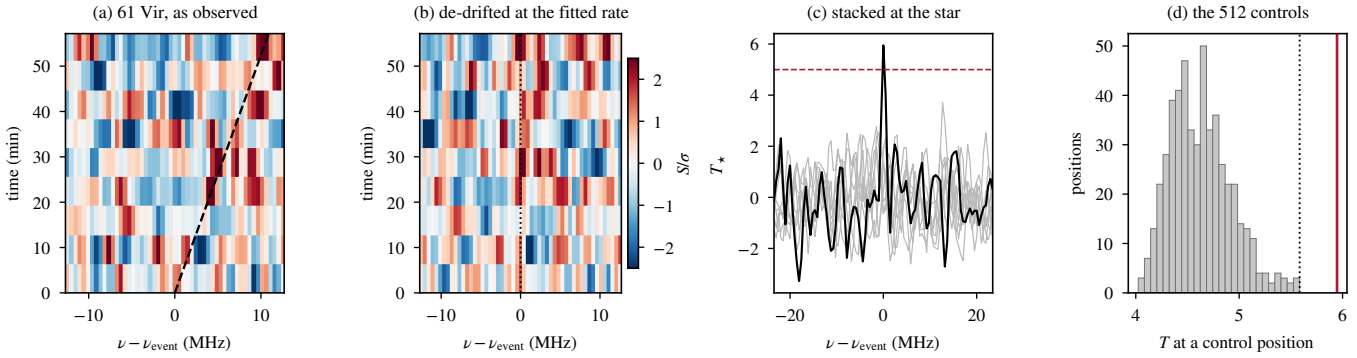


FIG. 4.— **The strongest unattributed crossing of the quality-passing data, and the control field it leads.** Frequencies are in the star's rest frame. *(a)* The dynamic spectrum at 61 Vir's position around the crossing at 346.3232 GHz, observed 2017 May 20, the fitted drift of $+3182\,\mathrm{Hz\,s^{-1}}$ (observatory frame; +3221 in the star's frame) dashed, 225 integrations binned into 9 and plotted in units of each bin's noise. *(b)* The same rows shifted by that drift. *(c)* The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise, the trigger dashed. *(d)* The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted).

4.5. *A stellar-frame stacked search*

Combining a star's epochs in its own rest frame deepens the limit, provided the emitter holds still in that frame. A frequency fixed in the stellar frame does not stay fixed at the telescope, since the barycentric term alone swings by a median $8.0\,\mathrm{km\,s^{-1}}$ between the epochs of one star, many channels at the 488 kHz Class A width. Each epoch is therefore transformed to the stellar rest frame, divided by its own primary-beam response and combined with the others by inverse variance over the channels of common coverage. The transform is checked on HD 285968's line, which 4 epochs observe $12.2\,\mathrm{km\,s^{-1}}$ apart and place at one stellar-frame frequency to $0.03\,\mathrm{km\,s^{-1}}$. The criterion is applied in the same way at the stacked control positions, and every stacked limit is measured by injection into the stack.

Of the surveyed stars, 59 can be stacked, 5 of them with blocks beyond the census, and over 316 stacked groups the flux-density limit improves on each star's best single epoch by a median factor 1.50 and by 4.45 at most. A further 10 stars, all control stars beyond 40 pc, are stacked and left out of those figures. The gain is 0.79 of what $\sqrt{N}$ equal epochs would give, the shortfall being unequal epoch depth. In the stack, 2 groups pass the full criterion and both are known lines, the β Pictoris carbon monoxide of §4.3 in two transitions, so the positive control passes in the stack as well, and the survey's own $\pm 50\,\mathrm{km\,s^{-1}}$ mask leaves none of them unattributed.

Every stacked limit holds only for an emitter steady in the stellar frame to within one of its own channels. No crossing of §4.4 drifts slowly enough to survive a fine-channel stack, which admits far less stellar-frame excursion than a coarse one, so the single-epoch limits of §4.1 are the ones that apply generally. The deepest stacked limit is Proxima Centauri's, 1.9×10^{13} W from 25 census blocks of coarse 15.625 MHz channels, and like every Class B value it is a limit on an unresolved narrowband spectral excess. It also needs a qualification of its own. A transmitter on Proxima Centauri b would swing over $94 \sin i\,\mathrm{km\,s^{-1}}$ in its orbit (Anglada-Escudé et al. 2016; Kervella et al. 2017; Akeson et al. 2013), some 5 channels at $\sin i = 1$. The stack would cancel such a signal unless the orbit is seen nearly face-on, with $\sin i \leq 0.22$, which holds for 2.5 per cent of random orientations. Within one block, by contrast, the planet moves at most 6 per cent of a channel, so the host's best census single-epoch limit, 7.3×10^{13} W in a Class B window, is the one that applies to it.

5. DISCUSSION

An archival search can only be as good as the observations it inherits, so we ask three questions of the result: what new parameter space the archive allowed us to search, what the absence of a detection actually constrains, and what stops the search from saying more. The answer to the last one also says what a dedicated experiment should do differently.

5.1. *What was searched*

The parameter space opened here is millimetre and submillimetre frequency toward the nearest stars. The archive gives 47.7 GHz of sky frequency between 114 and 873 GHz toward systems within 40 pc. The two 203 GHz searches of nearby stars each looked at a single line. The only search that covered more, that of Mason et al. (2025), spanned 117 MHz of Band 3, and its targets were distant stars that happened to lie in calibrator fields (§1). Most of this range had therefore not been searched at all for narrowband emission from the direction of a nearby star.

The price of using an archive is that the channels were chosen by other observers, for other purposes. They range from 15.3 kHz to 31.25 MHz, while the centimetre-band carrier searches work with channels of about 2.8 Hz (Fig. 1b). A carrier much narrower than one of these channels is measured against the noise of the whole channel, and this is one reason why the limits reached here lie well above those of the centimetre surveys.

Union bandwidth also overstates what was searched, because different frequencies were observed toward different stars, and to very different depths. Figure 2c therefore counts, at each sky frequency and EIRP, the systems in which a carrier would have reached the threshold nine times in ten in a single epoch. At 1×10^{15} W the covered frequency falls to 19.1 GHz, 40 per cent of the union, and 25 systems are reached at some frequency, 18 of them near 230 GHz. However high the power, no frequency is covered toward more than 38 of the 61 systems, and

only above 2.9×10^{15} W is half of the sample reached at any single frequency. The search is thus deep toward a few systems at a few frequencies, and shallow or absent almost everywhere else.

The 1258 coarse-channel windows add a constraint of a weaker kind. Their channels are too wide to follow a drift, so they limit only an unresolved narrowband spectral excess, but they are numerous, and stacked in the star's own rest frame they can be deep. Toward Proxima Centauri, the nearest star, the stacked limit reaches 1.9×10^{13} W for an emitter that stays within one channel in the stellar frame (§4.5).

5.2. *What the non-detection constrains*

Every carrier limit in this paper applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while an archival block was on sky, and drifting linearly within its window's ceiling of 12.0–13.3 nHz, a fractional drift rate that corresponds to a line-of-sight acceleration of 3.60–4.00 m s^{-2}. That ceiling covers the orbital acceleration of every known planet in the sample except TRAPPIST–1 b, which is reached only as precisely as its host's mass and orbit are known (§3.1). A transmitter on a rapidly rotating planet, however, uses most of the ceiling on the planet's spin. The Class B limits are of a different kind again, on an unresolved spectral excess with an injection calibration of their own (§4.5).

A signal that departs from this model is constrained less, or not at all. Emission as wide as the widest line in these fields needs 29 times the power of a one-channel carrier to be found, and a carrier that is switched on for only part of a block is recovered at that fraction of its amplitude. Nothing is constrained inside the ± 50 km s^{-1} mask around each listed transition. Nor does the survey say anything about the stars within 40 pc that ALMA has not observed, which are almost all of the 17 566 that this volume holds.

There are also two blind spots of a more specific kind. The analysis forms (XX + YY)/2, which retains the full intensity of a circularly polarised carrier, so what is lost is not sensitivity but the discrimination that polarisation would offer against unpolarised astrophysical emission. ALMA's feeds are linear, and the circular signature lies in the cross-hand products, which this analysis does not extract. The other blind spot is historical. The 203 GHz positronium hyperfine line that Steffes & DeBoer (1994) and Mauersberger et al. (1996) looked for lies in ALMA Band 5, where this survey holds 4 coarse-channel windows, all toward α CMa B, and none of them covers it.

5.3. *What prevents stronger conclusions*

Depth is not the main limit. As an illustration, a 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D\nu/c)^2 = 5.9 \times 10^8$, and hence an EIRP of 5.9×10^{14} W. The gain grows as the square of the frequency, and the systems' best windows lie between 219 and 354 GHz, so the fair comparison is with the same dish tuned to each system's own best window, where its EIRP runs from 5.3×10^{14} to 1.4×10^{15} W. Tuned in that way it lies above the single-epoch limit of 24 of the 57 Class A systems that retain unmasked bandwidth and have a bounded position, so a transmitter of terrestrial scale would already have been within reach toward those systems. For comparison, the Arecibo planetary radar had an EIRP of about 2×10^{13} W at 2.38 GHz (Sheikh et al. 2025), about 60 times below the median limit, and 10 systems come within a factor of ten of it. Near that boundary HD 38858 (9.8 times) is inside it and GJ 3379 (10.1 times) just outside, although both limits print as 2.0×10^{14} W in Table F1, because the count is taken on the unrounded limits. The dish's beam, however, has a solid angle $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr and covers only 1.7×10^{-9} of the sky. EIRP is not transmitter power, and these limits apply to a beamed transmitter only if its beam happens to contain the Solar System. The archive is limited less by its depth than by the astrophysics that can imitate a carrier, and by the repeat coverage that it happens to hold.

At these wavelengths the false positive is line emission. A catalogued molecular transition lies within the mask for 23 of the 51 crossings in the census's quality-passing data. For 16 of them there is support beyond the frequency match: a second transition of the same species, catalogued gas on the sightline, or the crossing's own recurrence. The other seven rest on the coincidence alone, which is not an identification. None was identified as interference by the available diagnostics (Appendix E), which leaves 28 for the comparison with chance of §4.2.1.

Molecular lines also bring a test of their own that has no centimetre analogue. A line that fills the primary beam raises the star and its control annulus together, whereas a carrier from the star raises only the star. Of the 11 crossings whose statistic at the star exceeds that at every control position of their window, which is the test that singles out the field leader (§4.4), 10 are carbon monoxide toward β Pic and HD 48370 (§4.3). All of them are listed in `candidates/above_all_controls.csv`.

An archive supplies repeat epochs only by accident. A single crossing, however strong, becomes a testable claim only when the star is observed again in a separated session, at the same frequency for an emitter at rest in its frame or at the frequency a trajectory predicts for one that drifts, and the archive holds such repeats only where observers happened to return. We describe this by the same-frequency coverage $C_{\rm conf}(\nu)$, the number of systems toward which two separated blocks (Table 2) both cover the sky frequency ν. It measures coverage, and is not a completeness for an orbiting carrier, because a drifting carrier must also fall at its predicted frequency in the second block. Its support is 24.0 GHz, 50 per cent of the 47.7 GHz searched, and only 8.5 GHz beyond a year (Fig. 2b). Of the 61 systems, 40 have such a block and 13 have one more than a year away. The other 21 have none, for the two reasons set out in §6. Even where a repeat exists it can be too shallow to test anything, as it was for 6 of the 27 crossings read at a predicted cell.

5.4. *What a dedicated ALMA experiment should do differently*

Each of these limits has a remedy on the same telescope. A dedicated search can choose its targets and its frequencies instead of inheriting them. This archive favours dusty discs, which are the worst case for line confusion, and covers least the M dwarfs that host most

of the nearby habitable-zone planets. Fine channels are needed in every window, since the 1258 coarse-channel windows of the archive cannot carry a carrier search at all. Tunings placed away from the bright molecular lines recover most of the 3.05 GHz of Class A coverage that the mask costs here, and probably remove many of the false positives as well. Above all, the recurrence observations have to be scheduled deliberately, at the same tuning and days to years apart, so that every crossing meets a separated epoch at the frequency it predicts. This is chiefly a matter of scheduling, and each additional epoch tests a trajectory predicted independently from the others. Indeed 27 of the 61 systems already carry three epochs at a common frequency, but none of them has a crossing seen in more than one of those epochs. The obvious first target for such a programme is ALMA J153702653−33192492, the most discrepant of the objects whose repeat windows were compared with their own control positions (§4.4).

6. CONCLUSIONS

This paper asked whether any of the nearby stars that ALMA has observed shows a narrowband, drifting radio carrier of the kind an artificial transmitter would produce at millimetre and submillimetre wavelengths. Because the observations were taken for other purposes, it also asked how deep and how complete a search of the archive for such a carrier can be. To answer both we searched the archival ALMA spectra of 61 star systems within 40 pc. The fine-channel windows cover 47.7 GHz of sky frequency, in 26 separate intervals between 114 and 873 GHz. That union is 1364 times the 35 MHz of the wider of the two millimetre searches of nearby stars that preceded it, and the coverage of the median system alone is still 53 times larger. We removed the drift from each spectrum and compared it with control positions in the same field, so that the background at the star could be judged from the observation itself.

No transmitter was found. In 51 windows the stellar statistic reaches the threshold $T_\star \geq 5$. Of these, 23 coincide with catalogued molecular lines, 16 of them with further evidence of line emission, and none was identified as interference by the available diagnostics. The other 28 are consistent with chance: the control positions beside each star predict 24.0, with $P = 0.24$. The extraction field and the line mask could each have been chosen in two ways. Corrected for having been picked out of those four combinations, the smallest of their probabilities is 0.17, so no combination departs from chance at the 5 per cent level. Both probabilities are conditional on the line mask as adopted, which was completed after the crossings had been examined, and on the control positions being a fair model of the background at the star.

None of the unattributed crossings became a recurrent feature, although the molecular lines did, as they should. Wherever an unattributed crossing was observed again in a separated session and the repeat could be read, nothing was found at its own frequency in the star's frame. That shows only that none of them returns in the way a line would. A drifting carrier would by then have moved, and the archive can test one only by asking whether a single trajectory joins two crossings, which can tell crossings apart but can never confirm them. Recurrence is in any case necessary for an artificial origin but not sufficient: a feature found again would still need its line identity, interference and the behaviour of its frequency in the star's frame to be checked. For the crossings themselves, therefore, this is a non-detection, not an exclusion.

Because nothing was confirmed, what the search delivers is a set of single-epoch upper limits. In the 30 windows where an unattributed crossing, or a weaker unattributed cell, is already present, a carrier reaching only the threshold would not have stood out, so there the limit is quoted at the statistic actually observed and is an upper limit as well. That moves the limits of four systems, marked in Table F1, and leaves the median system limit and the count of systems at 10^{15} W unchanged. The limits apply to a carrier narrower than one channel, lying outside the molecular-line mask, radiating while ALMA observed and drifting by no more than 12.0–13.3 nHz, the drift ceiling discussed in §5.2. To reach the threshold nine times in ten in a single epoch, such a carrier needs an equivalent isotropic radiated power, or EIRP, with a median, over systems, of 1.2×10^{15} W in each system's most sensitive window. This is a single-epoch trigger completeness, not the chance of confirming a signal, since a crossing would still have to be found again. The deepest limit, 3.7×10^{13} W, is reached toward ϵ Eridani, and 25 of the 57 systems with a limit that does not depend on an unchecked position reach 1×10^{15} W.

It helps to set these numbers against something we have built ourselves. The Arecibo planetary radar, one of the most powerful transmitters ever operated on the Earth, had an EIRP of about 2×10^{13} W at 2.38 GHz. The median limit found here is about 60 times that, and even the deepest is 1.9 times it, so no system reaches an Arecibo-like transmitter, although 10 come within a factor of ten of it (counted on the unrounded limits, §5.3). The same comparison can be put as a distance. Because the received flux falls as the square of the distance, an Arecibo-like transmitter placed at a star of this survey would be detected only if the system were much closer to us than it is. For the median system it would have to be within 3.4 pc, and even in the most favourable case, TWA 3A, whose limit is deep for its distance, within 7.6 pc. With the atmospheric decorrelation divided out these become 3.0 pc and, for β Pic, the most favourable system on those values, 6.9 pc, although β Pic's value rests on an imputed decorrelation estimate. A 12 m dish radiating 1 MW is a more natural transmitter at these frequencies. Tuned to the frequency of each system's most sensitive window, between 219 and 354 GHz, its EIRP lies above the single-epoch limit of 24 of the 57 systems. It could only be seen, of course, if its beam happened to point at us, as discussed in §5.3.

These limits come from the single-epoch carrier search, and they are not the deepest number in the paper. Toward Proxima Centauri the Class B spectra, stacked in the star's own frame, reach 1.9×10^{13} W in 15.625 MHz channels (§4.5). That is a limit of a different kind. It applies to an unresolved spectral excess that holds its frequency in the stellar frame to within one of those wide channels over all the epochs combined, it carries its own injection calibration, and it cannot tell a carrier from a line. A transmitter on Proxima Centauri b would move across many channels in its orbit, and the stack would cancel it unless the orbit is seen nearly face-on, so for a drifting carrier the single-epoch limits remain the ones

that apply.

How complete the search can be is set more by the archive's coverage than by its depth. A crossing can be put to the test only if its star is observed again in a separated session, at the frequency the crossing predicts. Of the 61 systems, 20 were never observed in two separated sessions at all, so that nothing found toward them could have been confirmed at any power. For one more, separated sessions exist but share no covered frequency, so the same-frequency recurrence test cannot be applied there. A programme designed for the purpose, on the same telescope, would use fine channels throughout, tune away from the bright molecular lines and schedule its repeat observations deliberately, so that any crossing it found could be looked for again at the frequency it predicts (§5.4).

What, then, can be said about transmitters? A continuous, unresolved carrier at one of these stars, outside the line mask, drifting within the ceiling and radiating above the limit of the system's most sensitive window would have crossed the threshold nine times in ten in a single epoch. Such a transmitter is therefore excluded at the single-epoch trigger level, toward the systems and at the frequencies that the archive covers. A great deal remains possible. A transmitter below these limits would not have been seen, and neither would one that was switched off, or whose beam was not pointed at the Earth, while ALMA was looking. Signals at frequencies between the covered intervals or inside the line mask, drifting faster than the ceiling or broader than a channel, are not constrained at all. Nor is anything said about the nearby stars that ALMA has not observed, which are almost all of those within 40 pc.

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This paper makes use of the following ALMA data: the 67 projects listed in Table A3, each cited as `ADS/JAO.ALMA#` followed by its project code. The 407 primary-census execution blocks searched span 2013 October 6 to 2025 July 11, and we thank the principal investigators and teams who proposed and obtained them. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France); of the NASA Exoplanet Archive (Akeson et al. 2013), which is operated by the California Institute of Technology under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program; and of the Splatalogue spectral-line database (Remijan et al. 2007).

Use of generative artificial intelligence. Large language model assistance (Anthropic Claude) was used in manuscript preparation of this complex paper. Data analysis was done using the Python based framework ASTRA (White 2026, https://arxiv.org/abs/2610.36144) to assist in the archival retrieval and calibration scripts, search and vetting pipeline, statistical tests, and the generators that produce every number, and carefully cross-checked and validated. The scientific content, analysis, and conclusions were designed, executed, and verified by the authors. All errors and omissions are the responsibility of the author.

Testing and validation. No result here is accepted because a model reported it: every number in the text is written by the program that computes it from the data, and a separate audit re-derives the identities that must hold among the published values and refuses the paper where one does not. Every check in the analysis was first made to fail on deliberately wrong input before it was relied on.

Software. The pipeline depends on CASA 6.7.0 (CASA Team 2022), PYTHON 3 with ASTROPY (Astropy Collaboration et al. 2022), NUMPY (Harris et al. 2020) and SCIPY (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through ASTROQUERY (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

Author contributions. G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript; R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §3.1 and Appendix D, and agreed the dispositions of the crossing ledger.

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DATA AVAILABILITY

Every observation used here is public in the ALMA Science Archive under the 67 project codes of Table A3, and is retrieved by the execution-block identifier (Chen et al. 2022), which the crossing ledger gives for each of the 53 threshold crossings; the astrometry is Gaia DR3, the planet counts come from the NASA Exoplanet Archive, and the line mask from one Splatalogue query of CDMS and JPL made on 2026-10-06. The data and software are held in a versioned Zenodo record (White & Dey 2026). It contains the per-window catalogue of all 1663 windows, the Class A and Class B limits with the injection trials behind them, the crossing ledger, the control-field statistics, the execution-block and session accounting (every row of Table A1), and the resampling inputs and seeds. From these one program, `code/reproduce.py`, rebuilds the chance expectation and its probability, the completeness factor, the sensitivity summaries and the repeat-window comparison, and checks each against the value printed here; the principal tables also accompany the arXiv version as ancillary files. The record holds the

search pipeline as it ran, which takes a public execution block to the statistics of this paper.

Not every product was made by that final form of the pipeline. Two steps were repeated for the blocks that matter most, and they did not reach the same blocks. The first is recalibration with ALMA's final flags (§3), the second a fresh extraction and search with the final extractor on the recalibrated data. Of the 52 blocks that hold a crossing, set a system's best limit or come close to the threshold, 41 went through both steps. The other 11 could not be recalibrated with the disk and software available to us: seven older blocks would need their whole calibration replayed, two restore ALMA's flags only through an import several times larger than the raw data, one has flag tables that cannot be restored at all and one failed to calibrate on the import that fits the disk. Of these eleven, one holds a crossing (toward β Pic) and five set the best limits of six systems (HD 172555, η Crv, 61 Vir, PM J03433+1958, WD 0407+179 and LP 353-51), and the other five hold neither. The windows of all these blocks keep the products of the first forms of the extractor (`catalogues/re_extraction_status.csv`), and Table A2 sets this count beside that of the recalibration. The final extractor places the phase centre at the star's parallax-displaced position and keeps every field entry, which the first forms did not, and the very first also drew its control annulus differently. Those versions are archived with their settings, together with a table that gives, for every window, the versions that made its products; run again on freshly calibrated 7 m and 12 m blocks, they reproduce those products in every window tested. The dynamic spectra at the star and at its 8 retained control positions, around each of the 53 crossings, form a second file of the same record (180 MB). The calibrated visibilities, of order a terabyte, and the per-window spectra were not kept, for lack of storage, so this execution of the search can be run again from the archive with the same code but cannot be repeated exactly.

APPENDIX

A. SPECTRAL RESPONSE, RETENTION AND THE SURVEY LEDGER

A.1. *The response of a channel to a carrier*

A carrier seldom falls at the centre of a channel. ALMA's online Hanning smoothing (Cortes et al. 2026) leaves 0.50 of an unresolved carrier's flux in the peak channel at a channel centre and 0.42 on a boundary, and this loss enters the limits once, through the injections. Each synthetic carrier is deposited at a sub-channel phase drawn afresh and recovered through the unmodified trigger, so the measured completeness already contains it. The Class A factor separates into two parts, $2.99 \simeq 2.286 \times 1.309$. The first is $1/0.4375$, from the phase-averaged share of a carrier's flux that the 0.25/0.5/0.25 kernel places in the peak channel. The second is the margin that a peak channel needs above the nominal trigger for nine carriers in ten to return. $P_{\rm trig}$ and $S_{\rm min}$ therefore stay nominal, comparable with other programmes' thresholds, and the sensitivity carries the correction,

$$\mathrm{EIRP}_{90} = \frac{\kappa\,\rho_N}{R_{\rm px}\,\eta_{\rm smear}}\,4\pi d^2 S_{\rm min}\,\Delta\nu, \qquad S_{\rm min} = \frac{5\sigma}{A}, \tag{A1}$$

where $4\pi d^2 S_{\rm min}\Delta\nu$ is $P_{\rm trig}$, κ is the class factor, ρ_N the response ratio of the window's averaging mode, and A, $R_{\rm px}$ and $\eta_{\rm smear}$ are the primary-beam, parallax and smearing retentions. The noise σ is that of the time-stacked spectrum at the star, in each native channel. In every integration and channel it is the median absolute deviation over the control positions, scaled to a Gaussian standard deviation, and the integrations are then combined by inverse variance (`code/pipeline/search/seti_drift_search_generic.py`, lines 436–437 and 574).

The kernel is an approximation. A delivered channel is the mean of N Hann-windowed raw channels ($N = 1$ in 290 Class A windows, online averaging in 115), so carriers were also injected with each window's own response, carrier for carrier against the kernel, into 35 windows of every averaging mode. The ratio ρ_N of the two ninety-per-cent points is 0.91, 0.64 and 0.68 at $N = 1$, 2 and 4 in Class A, and 0.88 and 0.41 at $N = 1$ and 2 in Class B. Smearing within one integration, which the injections do not model, is divided out as $\eta_{\rm smear}$, the mean channel response over the sweep, and is never worse than $\times 1.05$.

A.2. *Primary-beam and positional retention*

The response applied, A in Eq. A1, is a Gaussian primary beam of the documented full width $1.13\,\lambda/D$ (Cortes et al. 2026) at each window's own dish diameter. Archival fields are pointed at catalogue J2000 positions, so these nearby, fast-moving stars sit off axis by a median of $0.082''$ and at most $17.99''$. At the furthest census offset the Gaussian agrees with the blocked-aperture response of CASA (CASA Team 2022) to 7.1 per cent. The statistic, ranks and attribution are ratios within one window, so A enters only $S_{\rm min}$. $R_{\rm px}$ is the modelled share of a carrier's amplitude that survives the annual parallax where the extraction omitted it, and the model errs on the safe side, since carriers injected at the full apparent position of the 8 furthest-out windows retain 0.92 and 0.58 where it predicts 0.89 and 0.49. ϵ Eridani has no parallax record, and its limit may be optimistic by up to 9 per cent.

Propagating a position also assumes straight-line motion. Of the 91 searched positions, the 90 census stars and HD 139084 B, 30 have a companion within 400 arcsec (Mason et al. 2001; Matson et al. 2026). The displacement, from a visual orbit, from the separation with masses from Pecaut & Mamajek (2013), or from the proper-motion anomaly (Kervella et al. 2022), moves no system's limit by more than 16 per cent. Barta 161 12, HD 155555C and LHS 1140 are different. Their astrometric solutions are disturbed, yet they have no known companion and no Hipparcos epoch, so the positional loss in their 24 windows has no bound and their limits hold only at the adopted positions. They are marked in Table F1 and left out of the median, the count at 10^{15} W and Fig. 3 (§4.1).

A.3. *The survey ledger*

Every headline count is read from Table A1. The data archive repeats the table row for row in `catalogues/block_ledger.csv`, and `catalogues/count_accounting.csv` itemises each of its rows under the same number and says which rows it is the sum of. The planet counts are from the NASA Exoplanet Archive `pscomppars` table (Akeson et al. 2013), accessed 2026 October 4, and Table A3 lists the ALMA proposals the searched blocks belong to.

Two counts of reprocessed blocks appear in the paper, and they are not of the same blocks. The recalibration with ALMA's final flags, described in §3, was planned on the blocks that then set a quoted limit or held a crossing. The fresh extraction with the final extractor was planned later, on the blocks that held a crossing, a best limit or a statistic close to the threshold in any set, as the Data availability statement describes. Table A2 sets the two side by side, block by block, and adds every block that holds a crossing or a system's best limit now.

B. THE CONTROL ASSUMPTION, THE HOLD-OUT AND THE RECURRENCE BOUND

The chance expectation of §4.2.1 predicts how often the star should cross the trigger from how often control positions in its own field do. It therefore rests on one assumption the searched data cannot test, that the stellar position behaves like a control position. Below the trigger it does not quite do so. In the band $4 \le T < 5$ the star carries 1.102 times a control position's above-threshold cells (1.05–1.16 at 95 per cent, blocks resampled). Rescaling the expectation by that ratio is a sensitivity check (§4.2.1); the control positions' own rate, the more demanding of the two, is kept for the count under test. For the 28 unattributed crossings to stand above chance at 5 per cent, stars would have to trigger at 0.82 times a control position's rate or less.

The stellar excess is not a primary-beam effect, since a noise rise toward the field edge would put the stellar rank above one half, the opposite of what is seen, and the probes' radial slope on the reserved blocks, -0.026 ± 0.024, is small and of the wrong sign. It appears only in the 7 m-array data, 1.15 (1.1–1.2) against 1.03 (0.94–1.12) on the 12 m array, and its cause has not been identified.

A calibration error that varies with frequency, such as a bandpass residual or a coarse system temperature interpolated across a telluric line (Hunter et al. 2018), would act in the sense observed: it multiplies the continuum, so it acts at the star and not on the empty sky around it. Such a residual matters even if it is too small to make a crossing by itself. A small error coupled to the continuum, added to noise already near the trigger, would still change how often the star crosses it, and the searched data cannot separate that from a real difference between star and controls. The probabilities therefore rest on the stellar-background model (§4.2.1).

B.1. *The reserved hold-out*

Because the statistic and the mask were set on the searched sample, 77 of 484 blocks (315 windows toward 36 stars) were held out by an identifier rule, $\mathrm{sha256}(\mathrm{UID})_{[:8]} \bmod 5 = 0$. Two guards, which also use identifiers alone, keep a block on a named list in the survey and ensure that no system loses every block. Their independence of the statistic and the mask is our assertion, which a reader cannot check. Since the rule hashes the block and not the session, 64 per cent of them also share calibrators, weather and setup with a searched block. They trigger in 6 of 62 windows against 6.5 predicted, and the 28 that share no session hold two trigger events against 1.9, so they neither confirm nor contradict the stellar ratio.

Counted in the way the census is, in unattributed crossings and with the same statistic, mask and screen, the 62 fine-channel windows of the hold-out give 4.3 expected by chance against 5 observed, with $P = 0.42$. The hold-out is never pooled with the census.

Whether the hold-out's crossings were looked at before the line list was completed cannot be told from the record. The rule that selects the species had been written down before the line query of 2026 October 6, and the hold-out crossings were listed in the same session, but the record does not say which came first. The query then applied that rule mechanically. No hold-out disposition depends on the later decision about SO, since every unattributed hold-out crossing lies at least $223\,\mathrm{km\,s^{-1}}$ from a listed transition.

B.2. *The recurrence bound*

A crossing is tested for a stationary recurrence by reading its predicted cell, one channel at the carried drift, in the later blocks that cover it. The bound below therefore applies only to an emitter at rest in the star's frame. With S the discovery amplitude and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ the inverse-variance combination of the repeat readings, the two signal-to-noise ratios are $T_{\mathrm{pers}} = S/\bar{\sigma}$ and $T_{\mathrm{rep}} = \bar{a}/\bar{\sigma}$. A discovery amplitude selected from 10^2 to 10^7 cells at $T_\star \simeq 5$ is biased high, so it is not compared with the repeat directly. The bound is instead the largest fraction of the discovery amplitude still allowed by the repeat readings,

$$f_{\mathrm{rep}} = \frac{T_{\mathrm{rep}} + 5}{T_{\mathrm{pers}}},$$

and T_{rep} is centred on zero under the null, scattering by 1.022 about -0.016 over 371 single-cell repeat readings. The reading is conditional. Given an assumed true amplitude, the bound recovers a carrier with probability $\Phi(T_{\mathrm{pers}} - 5)$ at the discovery amplitude and $\Phi(5T_{\mathrm{pers}}/T_\star - 5)$ at the nominal threshold-equivalent amplitude $5S/T_\star$, and the 6 crossings where these are smallest are reported as untested.

C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask decides which crossings those are.

The species list was assembled by hand. It holds the bright millimetre disc and cloud tracers the pipeline checked from its first search, with [C I] and hydrogen recombination added during the search and SO after a crossing was found on one of its lines. It was completed at the query below, after the crossings had been examined, and it omits other species seen in discs and clouds, such as CH_3OH, DCO^+ and N_2H^+. The mask contains every

TABLE A1

The survey ledger: (a) execution blocks and windows, hours being window-hours, so the classes partition the hours as they partition the windows; (b) events, dispositions and confirmation. Each sub-row is a subset of the row its number extends. Note d gives the rule by which the counts of panel (a) sum.

(a) Execution blocks and windows

	Set of execution blocks[d]	Blocks	Windows	Stars	Systems	Hours
1	Archival scope: blocks the query exposes	656	–	–	–	–
1.1	… processed with the same pipeline	479	–	–	–	–
1.2	… not processed at the freeze	177	–	–	–	–
2	Processed with the same pipeline, 5 of them outside the scope query	484	–	–	–	–
2.1	**Primary census**	407	1663	90	83	1178
2.1.1	Class A, fine channels[a]	–	405	66	61	240
2.1.2	Class B, coarse channels[a]	–	1258	89	82	938
2.2	Reserved hold-out	77	315	36	–	237
3	Not processed at the freeze	177	–	–	–	–
3.1	Calibrated and searched	144	–	–	–	–
3.1.1	**Archival extension**, reported	115	463	16	16	362
3.1.1.1	… of which Class A, fine channels	–	67	11	–	48
3.1.2	**Archival tail**, searched after the extension was frozen	8	32	5	5	23
3.1.2.1	… of which Class A, fine channels	–	14	5	–	10
3.1.3	ϵ Eri Band 6, withheld[b]	8	32	1	–	25
3.1.4	Searched, no window reached an export[c]	13	–	–	–	–
3.2	Not searched: 18 without a delivered calibration, 11 failed to calibrate, 3 not attempted, 1 too large for the working disk	33	–	–	–	–
4	Beyond 40 pc: out-of-sample control	47	198	22	–	87

[a]The two classes partition the census windows, not its blocks: 255 blocks carry a Class A window and 392 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

[b]Every ϵ Eri block of the extension is a Band 6 block, where the Gaussian primary-beam form used for the response does not hold, so the extension reports none of its windows. None of its 32 windows holds a crossing. [c]Calibrated and searched, but no window of these blocks reached any export, so they hold no crossing and set no limit; the count is what row 3.1 leaves after rows 3.1.1–3.1.3. [d]Each block, window and hour count is the sum of the rows numbered under it, except a row marked *of which*, which is a subset of its parent and is not summed. Stars and systems are not summed, because one star can be observed in several sets.

(b) Events, dispositions and confirmation

	Count	Value	What it counts
1	Threshold crossings at $T_\star \geq 5$	53	rows 1.1 and 1.2
1.1	… in the primary census	51	rows 1.1.1 and 1.1.2
1.1.1	Class A, fine channels	51	the carrier experiment
1.1.2	Class B, coarse channels	0	the released catalogue's 4, all in the one block that fails the quality criterion, fall below the trigger on recalibration; 0.12 expected at the 1254 other coarse windows' own control positions
1.2	… in the archival tail	2	both unattributed, toward HD 14055, in windows no expectation covers
2	Quality-passing crossings of the census	51	the same rows as 1.1.1
2.1	… attributed to a transition	23	of 23 coincidences inside the mask in the star's own frame; rows 2.1.1 and 2.1.2
2.1.1	… with evidence beyond the coincidence	16	recurrence at the predicted cell, a second transition of the same species, or catalogued gas on the sightline
2.1.2	… on the frequency coincidence alone	7	η Crv, CD−57 1054, CP−72 2713, HD 161868, HIP 10679 and HR 1010: consistent with molecular emission, not identified as it
2.2	… unattributed	28	rows 2.2.1 and 2.2.2
2.2.1	… in the primary census	28	against 24.0 expected by chance at the 405 windows' own control positions – a count chance delivers with probability 0.24
2.2.2	… coincident with an atmospheric transition	0	within 62.5 MHz of one in the topocentric frame (Appendix E), and counted as atmospheric on both sides of that comparison
2.2.3	… present again at its stellar-frame cell in a later block	0	none is a recurrent feature; 13 attributed ones are
3	Class A systems	61	of 83 surveyed; rows 3.1 and 3.2
3.1	… with same-frequency coverage in a separated session	**40**	**the only confirmation count quoted**: a later block in another session, among the census and the repeat blocks the recurrence test reads, covering a frequency the first covered; 13 of them over a year later. For an accelerating carrier, confirmation further requires a predicted frequency within the repeat coverage, with adequate sensitivity
3.2	… with no separated second block at a covered frequency	21	no confirmation possible by the same-frequency test over the 141 blocks of this archive the recurrence test reads; 36 of all 83
4	Class A union bandwidth	47.7 GHz	26 disjoint islands
4.1	… removed by the line mask	6.4 %	3.05 GHz: the figure the text quotes, against 4.3 per cent of the 118.1 GHz archival union
5	Threshold crossings in every searched set	75	rows 5.1 to 5.5
5.1	… in the primary census	51	row 1.1
5.1.1	… at HD 139084 B	1	in a catalogued census window, at a position the catalogue does not tabulate; on neither side of the comparison of row 2.2.1
5.2	… in the archival tail	2	row 1.2
5.3	… in the reserved hold-out	6	enters no census statistic
5.4	… in the archival extension	10	enters no census statistic
5.5	… beyond 40 pc	6	the out-of-sample control

TABLE A2
Recalibration and extractor of the blocks counted in §3 or the Data availability statement, or holding a crossing or a best limit now.

Block	Recalibrated[a] final	Recalibrated[a] older	Delivered calibration final extractor[b]	Delivered calibration older	All
Holds a crossing	45	0	2	1	48
Sets a system's best limit	42	0	0	8	50
Sets another quoted limit	54	0	0	2	56
None of these now	48	0	0	5	53
All	189	0	2	16	207
of which counted at the recalibration[c]	163	0	2	13	178
of which planned for reprocessing[d]	41	0	0	11	52

[a]With ALMA's final flags. [b]The final extractor or an older one made the products of every window of the block. [c]The blocks that set a quoted limit or held a crossing when the recalibration was planned. [d]The blocks that held a crossing, set a best limit or came close to the threshold when their reprocessing with the final extractor was planned. A block appears in a row by what it holds now, crossing first. The table is `catalogues/recalibration_reconciled.csv` in the data archive.

TABLE A3
The 67 ALMA proposals the searched blocks belong to, as ADS/JAO.ALMA#*code*. Every census block belongs to one.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.00645.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.00696.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

transition of these 12 species inside the 46 searched intervals with an upper-state energy below 150 K, 111 transitions in all (Table C1). Only main hyperfine components are included, with no strength cut, and all come from one Splatalogue query (Remijan et al. 2007) of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) dated 2026 October 6.

Attribution is decided in the star's own rest frame. A crossing's topocentric frequency is carried to the barycentre with its own block's pointing and mid-time, and then to the star with its systemic velocity, giving $\nu_\star$. The crossing is attributed if, for any listed transition of rest frequency ν_0,

$$|\Delta v_\star| \le v_\mathrm{h}, \qquad \Delta v_\star = c\,\frac{\nu_\star - \nu_0}{\nu_0}, \qquad v_\mathrm{h} = 50\,\mathrm{km\,s^{-1}},$$

so that $\Delta v_\star$ is positive for a crossing above the transition in frequency. The same test, in the same frame, decides whether a control maximum is attributed (Appendix D). Of the 51 quality-passing census crossings, 23 meet it, and these are the 23 attributions; the hold-out is treated identically.

The windows were placed by their observers for their own lines, and that limits what "unattributed" means. Of the Class A windows, 39 per cent are centred within v_h of a listed transition, against 1–8 per cent for the same windows displaced, so a crossing is unattributed against a list that these setups helped to shape.

As a check that changes no disposition, every crossing is also read in the local standard of rest. Toward ALMA J153702653−33192492, HD 139084 B, HD 285968 and HD 48370 the CO($1 \to 0$) survey of Dame & Thaddeus (2022); Dame et al. (2001) shows emission at the crossing's own LSR velocity (within 3 km s^{-1} of its peak, and at least 0.23 K), so 6 crossings toward these stars lie at the velocity of interstellar gas. Elsewhere the Planck CO map (Planck Collaboration et al. 2016) stays below 0.178 K km s^{-1}.

Keplerian motion where these molecules survive sets the half-width, $\lesssim$15 km s^{-1} beyond 10 au. It has a cost. Over the union of searched sky frequencies the mask removes 3.05 GHz of the 47.7 GHz Class A union, 6.4 per cent, and 5.04 GHz of 118.1 GHz over both classes, 4.3 per cent, although each limit is formed over its own window's channels outside the mask in its own star's frame (§3.2).

TABLE C1
The line mask, by species: the number N of transitions, the frequency range they span inside the searched intervals and their upper-state energies.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03 – 231.90	–
SO	15	100.03 – 682.68	16–143
H_2CO	13	101.33 – 491.97	10–106
$C^{18}O$	3	109.78 – 329.33	5–32
CN	50	113.12 – 680.30	5–114
CO	3	115.27 – 345.80	6–33
CS	3	195.95 – 342.88	24–66
SiO	2	217.10 – 347.33	31–75
^{13}CO	8	220.40 – 330.59	16–32
HCN	7	354.50 – 354.51	43–43
HCO^+	1	356.73 – 356.73	43
[C I]	1	492.16 – 492.16	24

D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

D.1. *The expectation and its resampling*

The comparison with chance asks whether the stars cross the trigger more often than empty sky in their own fields. Each window supplies its own answer to how often empty sky does so, from its control positions, and one attribution set serves both sides, the mask of Appendix C being applied to a control maximum exactly as to a crossing. Over the 405 Class A windows w of the primary census the expected number of unattributed crossings is

$$\bar{N} = \tau \sum_w r_w\,(1 - a_w), \qquad r_w = \frac{n_w}{N_\mathrm{ctrl}},$$

where n_w of the window's $N_\mathrm{ctrl} = 512$ control positions reach the trigger, a_w is the share of control exceedances that land inside the mask, and τ is the share the atmospheric screen leaves. The rates alone give 36.3 expected crossings.

The attributed share has to be measured where control maxima actually fall in frequency, and that needs their spectra. The extractor kept a dynamic spectrum for only

8 of the control positions in each window, to save storage, and it chose them when the products were written, as entries spaced evenly through its list of control positions. That list is in the order of the seeded random draw that placed the positions, so the choice depends on nothing in the data. Of the 286 exceedances at these positions, 89 lie inside the mask, and attribution therefore takes 31 per cent of the expectation. One pooled share is used for every window. Windows differ in how much of their band the mask covers, however, so the share was also measured separately for the windows above and below the median masked fraction. The two shares are 0.44 and 0.10, from 181 and 105 exceedances, and with them the expectation becomes 24.1, with $P = 0.25$, so the pooled share is kept. Measured window by window instead, with each window's share drawn towards the pooled value, the expectation is 22.1 and $P = 0.14$, and the conclusion is the same.

The screen (Appendix E) then removes 9 of the 197 exceedances that remain at these positions, leaving $\bar{N} = 24.0$ against 28 observed.

Near a band edge the pooled exceedance rate is $\times 2.464$ the interior rate, but one window, toward HD 48370, holds 59 per cent of the near-edge exceedances. Without it the ratio is $\times 1.020$ ($\times 1.181$ on a held-out half, $p = 0.24$), so no edge correction is applied.

Some data enter neither side of the comparison. The 14 Class A windows of the archival tail are left out, and so is the crossing toward HD 139084 B. That crossing is attributed and is one of the 51 census crossings, but its position has no catalogue row. It therefore has no control positions of its own on the expected side, and so it is not counted on the observed side either. The window it lies in is still summed once, through the control positions about HD 139084. Which statistic every crossing enters is listed in `statistics/event_membership.csv` of the data archive.

Windows share execution blocks, so the count is compared through block resampling. In each of $B = 200\,000$ resamplings the blocks are drawn with replacement, each window's rate is redrawn binomially from its own probes, and the attributed share and the screen are redrawn from their own counts, giving $\bar{N}_b$; then

$$P = \frac{1}{B}\sum_{b=1}^{B} \Pr\big(N \geq N_{\rm obs} \mid \bar{N}_b\big)\,,$$

with N a Poisson count and $N_{\rm obs} = 28$, gives $P = 0.24$, conditional on the control-position model (Appendix B). As a check, a pseudo-star read at one control position in every window of a block is no more dispersed than one read window by window (variance ratio 1.0) and reaches 28 with probability 0.22. The widened recurrence window of §4.4 is read in the same way over its 96 distinct windows.

The bound quoted in §4.2.1 reads the lower tail of the same average over resamplings. With k added to the mean $\bar{N}_b$ of every resampling,

$$P_{\leq}(k) = \frac{1}{B}\sum_{b=1}^{B} \Pr\big(N \leq N_{\rm obs} \mid \bar{N}_b + k\big)$$

is the probability, on the same control positions, of a count as low as the one observed, and it falls as k grows while P rises. It is 0.06 at $k = 14$ and drops to 0.04 for one more. So 14 is the largest additional mean number of primary crossings compatible with the observed count at the one-sided 5 per cent level, conditional on the mask and on the control-position model. It is not a confidence bound on a realised number of crossings. Both tails are tabulated against k in `statistics/chance_allowance_table.csv`.

The control positions of one window are not independent draws, since neighbouring positions see much the same noise. Their layout can be rebuilt exactly from the extractor's seeded random positions, and two positions less than a tenth of a synthesised beam apart are then found to be correlated at 0.77. The correlation falls to half that by $0.14\,\theta_{\rm syn}$, and beyond half a beam no binned value exceeds 0.01. The correlogram is given in `statistics/control_correlation.csv`.

To see what this correlation does to the comparison, each window's positions were also grouped into blocks, either square cells from half a beam to a few beams across or arcs of the annulus. The 93 windows that carry 50 per cent of $\bar{N}$ then keep a median of 246 effective positions out of 512, and never fewer than 43. Drawing each window's rate from its own blocks instead of binomially raises the part of the scatter in $\bar{N}$ due to the window rates from 0.16 to 0.21. That is small beside the 1.4 of the block resampling as a whole. With the correlation kept, P becomes 0.24 and the bound 14, so the binomial redraw is kept. Each window's effective number of positions is given in `statistics/control_correlation_windows.csv`.

D.2. *The choices made after the crossings were seen*

Two choices behind the comparison were settled after the crossings had been examined, and each could have been made the other way. The control field can be the whole primary beam or only the positions nearest the star, and the mask can include SO or leave it out. The comparison was repeated for each of the four combinations of field and mask, with the same resampling. Each combination was also read with the expected rate raised by the star's excess just below the trigger (Appendix B). That excess is a measured factor and was not chosen, so it is a check on the background. Table D1 gives all eight readings.

The smallest probability in the table is 0.042, for the nearest field without SO. Multiplied by four, a Bonferroni factor that allows for its having been picked out of the table, it becomes 0.17, so no combination departs from chance at the 5 per cent level. Only the field and the mask count towards that factor. The stellar rate is one measured factor, 1.102, the same in every combination, and raising the expected count can only raise the probability. The smallest value therefore always lies among the four combinations read at the control rate.

D.3. *Execution-block quality*

The comparison assumes that each block's control ensemble behaves like noise, and in one block all 512 probes exceed the trigger. Every block is therefore judged on the median of its windows' control ensembles over its class median. The same measure, applied window by window,

TABLE D1
Unattributed crossings observed and expected, with the chance probability, for each control field and line mask.

field	mask	$N_{\rm obs}$	$\bar{N}$	P	$\bar{N} \times 1.102$	P
whole beam	with SO (adopted)	28	24.0	0.24	26.4	0.41
whole beam	without SO	31	24.9	0.14	27.4	0.28
nearest field	with SO	31	23.3	0.08	25.7	0.18
nearest field	without SO	34	24.2	0.04	26.7	0.11

also flags 5 of the 1663 windows at twice the survey median. That is a flag on single windows inside blocks that pass, and it is why the crossing toward HD 48370 is in quality-passing data and yet is withheld from the tests that depend on amplitude (§4.2). The criterion was written after the condition had been seen, but the choice is not a delicate one. The failing block sits at 4.1 and the next at 1.12, and its 4 coarse-channel windows hold no census crossing once ALMA's final flags are restored (§3).

D.4. *Registration*

Between epochs, the predicted cell is registered to 0.75 of a channel in every window. Widening each reading to the worst cell within 0.45 km s^{-1}, re-anchoring it along the discovery track, or raising the bar to 4.0 changes no disposition, although registration to a single channel is tested only on the unresolved line of HD 285968, which registers to 0.06 of a channel.

Where the epochs of a pair of crossings (§4.4) are close enough for the predicted offset to be tighter than the searched window, all 5 such pairs miss a drift changing at a constant rate by at least 3.6 times the resolution.

D.5. *The drift grid's edges*

The drift trials run from $-\dot{\nu}_{\rm max}$ in half-channel steps to the first trial at or beyond $+\dot{\nu}_{\rm max}$, so the searched interval is asymmetric by up to one step. Of the 59 crossings of the census, the archival tail and the hold-out, 53 have a fitted drift, and of these the 2 toward HD 161868 and TRAPPIST-1 lie on an outermost trial of their own window's grid. Their drift rates, in the observatory frame, and the accelerations they imply are lower bounds. Of these 2, 2 carry no amplitude bound.

E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, although an interferometer is a poor receiver of it, since a signal not in the far field at the phase centre is suppressed by its rapidly varying geometric phase. Allocations exist across the searched bands (International Telecommunication Union 2024), and the one documented hazard, spaceborne cloud radar at 94.0–94.1 GHz (Stephens et al. 2002; Wehr et al. 2023; Emerson 2004), overlaps 1 of the 1663 census windows, with no crossings.

The correlator flagging was not retained, so no per-channel interference flag exists, and interference can be looked for only in the way the crossings are distributed. The tests below therefore describe the population. They clear no single crossing, and they do not substitute for the ON/OFF cadence of a dedicated search. Each test could have failed, and none did.

Each p value is computed against the same peaks redrawn on their own channel grids. Crossings cluster in sky frequency only on the survey's own lines, with 15 of the 55 crossings of the per-window catalogue within ± 50 km s^{-1} of a searched transition against 6 expected ($p < 3 \times 10^{-4}$). The 30 unattributed quality-passing crossings of the census do not crowd on one another between 1 and 300 MHz ($p = 0.3$), whereas the attributed ones do. None of the 59 crossings of the census, the archival tail and the hold-out lies within 62.5 MHz of a telluric line. Peaks are spread uniformly within their windows (median 0.51), which a fixed intermediate frequency would not allow.

Interference reaching several stars would make them cross at the same frequency. Coincidences between different stars within 5 MHz number 569 against 550 expected ($p = 0.1$). Of the 28 unattributed crossings in catalogued blocks, 19 have no catalogued transition in the image band and 4 have one that their own block saw too faint to leak; for 5 the image frequency was not observed.

The one evident concentration is at the carbon monoxide tunings. The 16 crossings between 344.2 and 346.4 GHz include 10 of the 33 unattributed crossings of the census and the hold-out, against 10 expected from where the windows were tuned.

F. PER-SYSTEM SENSITIVITY

Table F1 gives for each system the limit of its best window outside the attribution mask (§3.2). A bound pair appears under its primary's name, and the limits at both ends of the calibration interval are in `sensitivity/system_limits.csv` of the data archive (White & Dey 2026). Beside each limit are that window's centre frequency ν and channel width $\Delta\nu$, the system's merged coverage $\Delta\nu_{\rm cov}$, the percentage *dec.* by which atmospheric decorrelation leaves the limit optimistic, and whether blocks of different sessions cover a common frequency more than a day (d) or a year (yr) apart, over the census and the 141 blocks the recurrence test reads. A dagger marks the 7 systems whose best window was itself injected into, 6 of them among the 57 systems with a bounded position. The other 51 limits of those 57 are transferred from the class average. For the 45 of them whose best window lies inside the configurations the injections covered, a single limit is uncertain by the dispersion between injected windows, $\times 0.81$ to $\times 1.12$. A pilcrow (¶) marks the 6 systems whose best window lies outside them, 51 Eri, ALMA J153702653-33192492, HD 172555, PM J03433+1958, TWA 7 and WD 0407+179. No measured dispersion applies to those limits, so we also give the summaries without them: over the other 51 systems the median limit is 1.1×10^{15} W and 25 reach 10^{15} W.

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TABLE F1

Per-system sensitivity of the carrier search. For each of the 61 systems the fine-channel search covers, the $EIRP_{90}$ of its best window, retentions applied, in order of the limit down the left half and then down the right half. Columns and marks: Appendix F and the notes below.

System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.	System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.
ϵ Eri	0.37	229.98	977	2.3	–	yr	κ CrB	14†	230.57	977	1.7	2*	–
LP 876-10	0.74	345.83	488	1.7	4*	d	HD 170773	15	230.13	488	7.4	2*	d
τ Cet	0.79	230.55	488	1.8	0.4*	d	HD 61005	17	230.52	244	19	3*	d
HD 69830	0.94†	335.38	244	3.8	0.5	d	LP 736-15	17	345.82	977	1.7	4*	–
AU Mic	1.1‖	230.56	488	3.5	3*	yr	LHS 1140	18§‖	230.00	488	1.8	0.4*	d
61 Vir	1.1	345.79	488	1.8	5*	yr	HD 172555	19¶	340.94	488	5.3	200	–
TRAPPIST-1	1.5	346.58	488	1.7	5	d	HD 155555C	19†§	346.01	488	1.7	43	–
AT Mic AB	1.5	346.04	488	1.7	3	–	L 836-122	19	346.00	488	1.7	1	–
β Pic	1.6‡	226.85	122	4.4	2*	yr	TWA 7	20¶	345.99	244	0.87	26*	d
HD 38858	2.0	230.50	488	1.8	3*	yr	GJ 2006A	22	346.02	488	1.7	1	–
GJ 3379	2.0‖	229.96	488	1.9	0.4*	d	CD−57 1054	22	345.99	488	1.7	20	d
HD 207129	2.2	230.57	488	1.8	3*	yr	HD 14055	22	230.11	488	7.2	3	yr
GJ 14	2.8	227.22	488	3.5	19	d	CP−72 2713	22‖	345.12	488	5.8	64	d
HD 139664	2.9	230.55	488	1.7	3*	–	HD 23484	24	230.53	977	1.8	0.2	d
HD 202628	3.4	230.51	488	1.8	4*	yr	51 Eri	25¶	230.53	1953	1.7	11*	–
HD 53143	3.4	230.53	977	1.8	2*	yr	TYC 2211-1309-1	26	346.00	488	1.7	5	–
HN Lib	3.5†	229.99	488	1.8	0.4*	d	UCAC3 176-23654	26	346.02	488	1.7	4	–
HD 10647	3.5	230.54	488	6.1	4*	yr	PM J03433+1958	28¶	230.13	488	1.7	2*	–
η Crv	4.0	343.35	488	7.2	27	yr	UCAC2 20312880	29	345.96	488	1.7	11	–
TWA 3A	4.7†	230.54	122	0.44	35*	d	LP 353-51	30	346.02	488	1.7	26	–
GJ 849	6.1	230.02	488	1.9	0.4*	d	Barta 161 12	31§	346.02	488	1.7	4*	–
HD 92945	7.3	355.11	488	3.7	6*	d	HD 139084	49	346.01	488	1.7	3	–
TX PsA	7.5	346.03	488	1.7	7	–	HD 45184	58	230.53	977	1.7	2	d
HD 285968	8.2	229.97	488	1.8	0.4*	d	WD 0407+179	76¶	230.13	488	1.7	2*	–
HD 161868	9.0	227.22	488	7.4	5	d	HD 48370	79†	230.73	122	3.3	0.4*	yr
HD 110411	11	219.58	244	4.7	2*	yr	HD 31392	82	230.52	977	1.7	0.9	d
HR 1010	12	230.53	977	2.8	1	d	ALMA J153702653-33192492	89¶	345.98	488	14	5*	d
HIP 490	12†	230.53	244	0.86	3*	–	HD 89452	160	230.53	977	1.7	0.7	d
HD 107146	12	355.14	488	3.7	6*	d	HD 377	250	230.54	977	1.7	68	d
Wolf 219	13	230.13	488	1.7	2*	d	HIP 10679	–	–	–	0.11	–	–
LP 476-207	13	346.02	488	1.7	4	–							

Notes. A double dagger marks a row keeping less than half its channels outside the mask; HIP 10679, whose 2 windows both lie inside it, has dashes. A section mark (§) flags Barta 161 12, HD 155555C and LHS 1140, whose limits hold only at the adopted position (Appendix A.2) and which are left out of the median and of the counts quoted over systems. *dec.* is a floor, the phase rms being recorded on the calibrator; it is measured where the delivery records one and imputed from blocks matched within a factor of two in baseline length where it does not (asterisk), and a dash marks the one system with neither. For the 12 systems where decorrelation exceeds ten per cent, the section-marked HD 155555C among them, the limits with it divided out are, in units of 10^{14} W, GJ 14 3.4, η Crv 5.1, TWA 3A 6.4, HD 172555 57, HD 155555C 28, TWA 7 26, CD−57 1054 26, CP−72 2713 36, 51 Eri 28, UCAC2 20312880 33, LP 353-51 37 and HD 377 430. The uncorrected value stays the one adopted, because the decorrelation is a floor. A double bar (‖) marks a system whose best-window limit is moved by the reading at the crossing's own level (§4.1). In a window that holds an unattributed crossing, a carrier reaching only the trigger would not have stood out, so the limit there is the power at which nine injected carriers in ten reach the crossing's own statistic. It is read from the pooled injections, since none of these windows was injected, and for AU Mic that window then ceases to be the best one.